

# $p$ -ADIC HEIGHTS ON SHIMURA CURVES AND HIGHER $p$ -ADIC GREEN FUNCTIONS

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## 1. INTRODUCTION

### 2. FILTERED $(\varphi, N)$ -MODULES AND GALOIS REPRESENTATIONS

Before delving into the  $p$ -adic aspects of Shimura curves and their cohomology, we will start recalling some basic fact about  $p$ -adic Galois representations and the associated category of abstract linear algebra objects introduced by [Fon79]. This will help us set up the notation and already establish the categories where the realizations of certain motives will live. We will follow the exposition of [Nek93].

Let  $F$  be a finite extension of  $\mathbb{Q}_p$  with ring of integers  $\mathcal{O}_F$  and residue field  $k$ . Denote by  $F_0 \subseteq F$  the maximal unramified subextension that can be identified with the fraction field of the ring of Witt vectors  $W(k)$ . Let  $\sigma : F_0 \rightarrow F_0$  the absolute Frobenius automorphism. Write  $G_F$  for the absolute Galois group  $G_F = \text{Gal}(\bar{F}/F)$ .

**Definition 2.1.** *Define the category  $\text{MF}_F(\varphi, N)$  of filtered  $(\varphi, N)$ -modules over  $F$  to be the category of finite dimensional  $F_0$ -vector spaces equipped with:*

- a  $\sigma$ -semilinear bijective map  $\varphi : D \rightarrow D$  called Frobenius,
- a  $F_0$ -linear map  $N : D \rightarrow D$  satisfying  $N\varphi = p\varphi N$  called monodromy operator,
- a decreasing filtration  $\text{Fil}^i D_F$  on the vector space  $D_F = D \otimes_{F_0} F$  satisfying  $\bigcup \text{Fil}^i D_F = D_F$  and  $\bigcap \text{Fil}^i D_F = 0$ .

The morphisms are defined to be compatible with  $\varphi, N$  and  $\text{Fil}^\bullet$ .

The category  $\text{MF}_F(\varphi, N)$  is an additive tensor category admitting kernels and cokernels. Given a filtered  $(\varphi, N)$ -module  $D = (D, \text{Fil}_D^\bullet, \varphi_D, N_D)$  we can define its  $j$ -th Tate twist to be the object

$$(2.1) \quad D[j] = (D, \text{Fil}_D^{\bullet-j}, p^j \varphi_D, N_D).$$

Consider  $D$  a filtered  $(\varphi, N)$ -module. For a rational number  $\lambda = n/m$  with  $n, m \in \mathbb{Z}$  and  $(n, m) = 1$ , denote  $D_\lambda$  the largest subspace of  $D$  which has an  $\mathcal{O}_{F_0}$ -stable lattice  $M$  satisfying

$$(2.2) \quad \varphi^m(M) = p^n(M).$$

The subspace  $D_\lambda$  is called *isotypical component* of  $D$  of slope  $\lambda$ . By definition of the monodromy operator we have  $N(D_\lambda) \subseteq D_{\lambda-1}$  for all  $\lambda \in \mathbb{Q}$ .

A  $p$ -adic representation  $V$  of  $G_F$  is a finite dimensional vector space over  $\mathbb{Q}_p$  endowed with a continuous linear action of  $G_F$ . We denote by  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  the category of such representations with obvious morphisms. It is an abelian tensor category with twists given by tensoring with Tate representation  $\mathbb{Q}_p(1)$ .

Fontaine [Fon94] constructed topological rings endowed with continuous  $G_F$  action

$$(2.3) \quad B_{\text{cris}} \subset B_{\text{st}} \subset B_{\text{dR}}$$

and certain extra structure. The embedding  $B_{\text{st}} \hookrightarrow B_{\text{dR}}$  depends on the choice of a branch of a  $p$ -adic logarithm.

Consider the following functors

$$(2.4) \quad \begin{aligned} D_{\text{cris}} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{MF}_F(\varphi, N) \\ V &\mapsto D_{\text{cris}}(V) := (V \otimes B_{\text{cris}})^{G_F}, \end{aligned}$$

$$(2.5) \quad \begin{aligned} D_{\text{st}} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{MF}_F(\varphi, N) \\ V &\mapsto D_{\text{st}}(V) := (V \otimes B_{\text{st}})^{G_F}, \end{aligned}$$

$$(2.6) \quad \begin{aligned} \text{DR} : \underline{\text{Rep}}_{\mathbb{Q}_p}(G_F) &\rightarrow \text{FilVec}_F \\ V &\mapsto D_{\text{dR}}(V) := (V \otimes B_{\text{dR}})^{G_F} \end{aligned}$$

where  $\text{FilVec}_F$  denotes the category of *filtered  $F$ -vector spaces*.

**Definition 2.2.** A  $p$ -adic representation  $V$  is called

- crystalline if  $\dim_{F_0} D_{\text{cris}}(V) = \dim_{\mathbb{Q}_p}(V)$ ,
- semistable if  $\dim_{F_0} D_{\text{st}}(V) = \dim_{\mathbb{Q}_p}(V)$ ,
- de Rham if  $\dim_F \text{DR}(V) = \dim_{\mathbb{Q}_p}(V)$ ,

The category of crystalline (resp. semistable, resp. de Rham) representations, denoted  $\underline{\text{Rep}}_{\text{cris}}(G_F)$  (resp.  $\underline{\text{Rep}}_{\text{st}}(G_F)$ , resp.  $\underline{\text{Rep}}_{\text{dR}}(G_F)$ ) is the full subcategory of  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  of crystalline (resp. semistable, resp. de Rham) objects.

The Tate twist  $D(n)$  is given by

$$(2.7) \quad D(n) := D_{\text{cris}}(\mathbb{Q}_p(n)) = D_{\text{st}}(\mathbb{Q}_p(n)) = (F_0, \text{Fil}^{-n}, p^{-n}, 0)$$

with the filtration concentrated in degree  $[-n]$ . The de Rham realization consists in the filtered vector space  $\text{DR}(\mathbb{Q}_p(n)) = (F, \text{Fil}^{\bullet-n})$ . The effect of tensoring by the Tate twist in  $\underline{\text{Rep}}_{\mathbb{Q}_p}(G_F)$  corresponds to taking the tensor product by (2.7) in the category of filtered  $(\varphi, N)$ -modules. In particular, according to our notation, for every  $p$ -adic representation  $V$  we have the following relation:

$$(2.8) \quad D_{\text{st}}(V(n)) = D_{\text{st}}(V)[-n].$$

**Definition 2.3.** A filtered  $(\varphi, N)$ -module  $D$  is called *admissible* if is isomorphic to  $D_{\text{st}}(V)$  for some semistable representation  $V$  of  $G_K$ . The full subcategory of admissible filtered  $(\varphi, N)$ -modules is denoted  $\text{MF}_F^{\text{ad}}(\varphi, N)$ .

**Theorem 2.1** ([CF00]). The functor  $D_{\text{st}}$  gives an equivalence of categories between  $\underline{\text{Rep}}_{\text{st}}(G_F)$  and  $\text{MF}_F^{\text{ad}}(\varphi, N)$ , which is compatible with exact sequences, tensor product and duality.

This theorem will be fundamental in the construction of the  $p$ -adic height of pairing à la Nekovář since we can establish the following group isomorphism.

**Corollary 2.1.** *The equivalence of categories induces a canonical group isomorphism*

$$(2.9) \quad \mathrm{Ext}_{\mathrm{Rep}_{\mathrm{st}}(G_F)}^1(V, W) \cong \mathrm{Ext}_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)}^1(D_{\mathrm{st}}(V), D_{\mathrm{st}}(W))$$

for every  $V, W$  semistable  $p$ -adic representations of  $G_F$ .

One of the advantage of working with the linear algebraic category of filtered  $(\varphi, N)$ -modules is given by the characterization and explicit description that we can give of the extension group  $\mathrm{Ext}_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)}^1$ .

**Lemma 2.1.** *Let  $D$  be a filtered  $(\varphi, N)$ -module and assume  $N$  induces a bijection  $N(D_0) = D_{-1}$ . Then there is a canonical isomorphism*

$$(2.10) \quad \mathrm{Ext}_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)}^1(F_0, D) \cong D_F / \mathrm{Fil}^0 D_F$$

where  $F_0$  denotes the filtered  $(\varphi, N)$ -module  $D_{\mathrm{st}}(\mathbb{Q}_p) = (F_0, \mathrm{Fil}^0, \mathrm{Id}, 0)$ . The map is often referred to as the Bloch-Kato isomorphism.

*Proof.* See [IS03, Lemma 2.1], [Mas11, Lemma 4.6]. Note the difference by a Tate shift  $[n+1]$ .  $\square$

It will be fundamental for our later computations to have a precise and explicit description of the isomorphism. We here recall the fundamental construction contained in the proof. Consider

$$(2.11) \quad e : 0 \rightarrow D \xrightarrow{\iota} E \xrightarrow{p} F_0 \rightarrow 0$$

an extension of filtered  $(\varphi, N)$ -modules whose class belong to  $\mathrm{Ext}_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)}^1(F_0, D)$  and satisfies the hypothesis of the Lemma. We have a decomposition of  $E_F \cong D_F \oplus F$  compatible with filtration

$$(2.12) \quad \mathrm{Fil}^0 E_F \cong \mathrm{Fil}^0 D_F \oplus F, \quad \mathrm{Fil}^i E_F \cong \mathrm{Fil}^i D_F \quad \text{for } i \neq 0$$

and there exists a splitting of  $p$  compatible with Frobenius and monodromy

$$(2.13) \quad s : F_0 \rightarrow E_0$$

where  $E_0$  is the slope-0 part. We then define  $\pi_{D_F}(s(1)) + \mathrm{Fil}^0 D_F \in D_F / \mathrm{Fil}^0 D_F$  the element associated to the class  $[e]$  of (2.11), where  $\pi_{D_F}$  denotes the projection to  $D_F$  under the decomposition  $E_F \cong D_F \oplus F$ .

**Remark 2.1.** *Lemma 2.1 fundamentally states that the class of a simple extension  $[e]$  is characterized by the comparison between the Frobenius-monodromy data and the filtration of the vector spaces.*

We will also need to characterize the extension groups for other filtered  $(\varphi, N)$ -modules not necessarily satisfying the hypothesis of Lemma 2.1. Consider the following exact sequence of admissible  $(\varphi, N)$ -modules

$$(2.14) \quad 0 \rightarrow F_0[-1] \rightarrow E \rightarrow V \rightarrow 0$$

where  $V$  satisfies the hypothesis above and  $F_0[-1]$  denotes the filtered  $(\varphi, N)$ -module  $D_{\mathrm{st}}(\mathbb{Q}_p(1))$ . Assume furthermore that  $V$  satisfies the following property

$$(2.15) \quad V^{\varphi=1, N=0} = (V^\vee[-1])^{\varphi=1, N=0} = 0.$$

I realized I needed the following lemma in order to compute one of the extension classes explicitly. It's better if you can double check that the proof works smoothly.

**Lemma 2.2.** *Let  $E$  be a filtered  $(\varphi, N)$ -module defined by (2.14). Then there is a canonical isomorphism*

$$(2.16) \quad \mathrm{Ext}_{\mathrm{MF}_F^{\mathrm{ad}}(\varphi, N)}^1(F_0, E) \cong E_F / \mathrm{Fil}^0 E_F \oplus F_0$$

where  $F$  denotes the filtered  $(\varphi, N)$ -module  $D_{\mathrm{st}}(\mathbb{Q}_p)$  as before. The map is often referred to as the Bloch-Kato isomorphism.

*Proof.* We consider the complex  $C_{\text{st}}^\bullet(E)$  defined by

$$(2.17) \quad C_{\text{st}}^\bullet(E) = [E \xrightarrow{(-i, \varphi - \text{Id}, N)} E_F / \text{Fil}^0 E_F \oplus E \oplus E[1] \xrightarrow{(0, N, \text{Id} - \varphi)} E[1]].$$

In [Nek93, Sec. 1.19], we have an isomorphism

$$(2.18) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, E) \cong H^1(C_{\text{st}}^\bullet(E)).$$

Explicitly, let

$$(2.19) \quad 0 \rightarrow E \rightarrow D \rightarrow F_0 \rightarrow 0$$

be an extension of admissible  $(\varphi, N)$ -modules whose class belongs to  $\text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, E)$ . Then we have a commutative diagram of the form

$$(2.20) \quad \begin{array}{ccccccc} 0 & \longrightarrow & E_F & \longrightarrow & D_F & \longrightarrow & F \longrightarrow 0 \\ & & \downarrow & & \downarrow \pi & & \downarrow \\ 0 & \longrightarrow & E_F / \text{Fil}^0 E_F & \xrightarrow{\sim} & D_F / \text{Fil}^0 D_F & \longrightarrow & 0 \longrightarrow 0. \end{array}$$

Pick a  $F_0$ -section  $s : F_0 \rightarrow D$  and define a triple

$$(2.21) \quad [-\pi \circ i(s(1), (\varphi - \text{Id})s(1), Ns(1))] \in E_F / \text{Fil}^0 E_F \oplus E \oplus E[1]$$

whose cohomology class  $\rho(D) \in H^1(C_{\text{st}}^\bullet(E))$  is independent of the choice of the section  $s$ . We now need to conclude that  $H^1(C_{\text{st}}^\bullet(E)) \cong \mathbb{Q}_p \oplus E_F / \text{Fil}^0 E_F$ . Consider the dual shifted sequence of (2.14)

$$(2.22) \quad 0 \rightarrow V^\vee[-1] \rightarrow E^\vee[-1] \rightarrow F_0 \rightarrow 0.$$

Under the assumption (2.15), there exists a unique  $(\varphi, N)$ -equivariant decomposition of  $E^\vee[-1]$  and then a decomposition

$$(2.23) \quad E = F_0[-1] \oplus V.$$

We then have that

$$(2.24) \quad \text{Im}(E \xrightarrow{((\varphi - \text{Id}), N)} E \oplus E[1]) = F_0[-1] \oplus \text{Im}(V \xrightarrow{((\varphi - \text{Id}), N)} V \oplus V[1])$$

since  $\varphi$  and  $N$  act respectively by the identity and trivially on  $F_0$ . On the other hand,

$$(2.25) \quad \ker(E \oplus E[1] \xrightarrow{(N, \text{Id} - \varphi)} E \oplus E[1]) = F_0[-1] \oplus F_0 \oplus \ker(V \oplus V[1] \xrightarrow{(N, \text{Id} - \varphi)} V \oplus V[1]).$$

We can then express the cohomology group  $H^1(C_{\text{st}}^\bullet)$  as

$$(2.26) \quad H^1(C_{\text{st}}^\bullet(E)) = E_F / \text{Fil}^0 E_F \oplus F_0 \oplus \frac{\ker(V \oplus V[1] \xrightarrow{(N, \text{Id} - \varphi)} V \oplus V[1])}{\text{Im}(V \xrightarrow{((\varphi - \text{Id}), N)} V \oplus V[1])}.$$

Recall that under the assumption (2.15), every simple extension of  $V$  factors through a filtered  $(\varphi, N)$ -equivariant map as a direct sum  $V \oplus F_0$ . In particular, we have

$$(2.27) \quad \text{Ext}_{\text{MF}_F^{\text{ad}}(\varphi, N)}^1(F_0, V) \cong H^1(C_{\text{st}}^\bullet(V)) \cong V_F / \text{Fil}^0 V_F.$$

From this we conclude  $H^1(C_{\text{st}}^\bullet(E)) = E_F / \text{Fil}^0 E_F \oplus F_0$ .  $\square$

The explicit description of the *Bloch-Kato isomorphism* (2.16) is given as follows. First of all fix the unique  $(\varphi, N)$ -equivariant decomposition  $E = F_0[-1] \oplus V$ . Then for every extension

$$(2.28) \quad 0 \rightarrow E \rightarrow D \rightarrow F_0 \rightarrow 0$$

consider the  $\varphi$ -equivariant splitting  $s : F_0 \rightarrow D$  inducing a  $\varphi$ -equivariant decomposition

$$(2.29) \quad D \cong F_0[-1] \oplus V \oplus F_0.$$

Fix a decomposition  $D_F \cong E_F \oplus F$  as in the previous case (2.12) then we define

$$(2.30) \quad (\pi_{E_F}(s(1)), \pi_{F_0[-1]}(Ns(1))) \in E_F / \text{Fil}^0 E_F \oplus F_0$$

where  $\pi_{E_F}$  denotes the projection under the decomposition  $D_F \cong E_F \oplus F$ , while  $\pi_{F_0}$  denotes the projection under the decomposition (2.29).

### 3. SHIMURA CURVES AND $p$ -ADIC UNIFORMIZATION

**3.1. The Shimura curve.** Let  $N$  be a positive integer that can be factored as  $N = pN^-N^+$  with

- $p$  prime;
- $N^-$  positive squarefree integer with odd number of prime divisors;
- $N^+$  positive integer relatively prime to  $pN^-$ .

Let  $\mathcal{B}$  be the indefinite rational quaternion algebra of discriminant  $pN^-$ . Fix  $\mathcal{R}^{\max} \subset \mathcal{B}$  a maximal order and  $\mathcal{R} \subset \mathcal{R}^{\max}$  an Eichler order of level  $N^+$ .

**Definition 3.1.** Let  $S$  be a  $\mathbb{Q}$ -scheme. An abelian surface with quaternionic multiplication (QM) by  $\mathcal{R}^{\max}$ , and level  $N^+$  structure over  $S$  is a triple  $(\mathcal{A}, i, C)$  with

- (i)  $\mathcal{A}$  is a principally polarized abelian scheme over  $S$  of relative dimension 2;
- (ii)  $i : \mathcal{R}^{\max} \hookrightarrow \text{End}_S(\mathcal{A})$  is an injective ring homomorphism;
- (iii)  $C$  is a subgroup scheme of  $\mathcal{A}$  which is étale locally isomorphic to  $(\mathbb{Z}/N^+\mathbb{Z})^2$  stable and locally cyclic under the action of  $\mathcal{R}$ .

**Definition 3.2.** The Shimura curve  $X := X_{N^+, pN^-}/\mathbb{Q}$  is the coarse moduli scheme representing the moduli problem over  $\mathbb{Q}$

$$\underline{\text{Sch}}/\mathbb{Q} \rightarrow \underline{\text{Sets}}$$

$$S \mapsto \{\text{Abelian surfaces with QM by } \mathcal{R}^{\max} \text{ and level } N^+ \text{ structure over } S\} / \sim.$$

$X$  is a smooth, projective, and geometrically connected curve over  $\mathbb{Q}$ .

Let  $B$  be the definite rational quaternion algebra of discriminant  $N^-$ , let  $R$  be an Eichler  $\mathbb{Z}[1/p]$ -order of level  $N^+$  in  $B$ . Define the group  $\Gamma$  to be

$$\Gamma := \{x \in R^\times : \text{nrd}(x) = 1\}.$$

Since the quaternion algebra  $B$  splits at  $p$ , fix an isomorphism

$$\iota_p : B_p = B \otimes_{\mathbb{Q}} \mathbb{Q}_p \xrightarrow{\sim} M_2(\mathbb{Q}_p).$$

The isomorphism  $\iota_p$  identifies  $\Gamma$  with a discrete cocompact subgroup of  $SL_2(\mathbb{Q}_p)$ .

**Theorem 3.1** (Čerednik–Drinfeld). *There is an isomorphism of rigid analytic varieties over  $\mathbb{Q}_{p^2}$ .*

$$(X_{\mathbb{Q}_{p^2}})^{\text{an}} \cong X_\Gamma := \Gamma \backslash \mathcal{H}_p.$$

In order to proceed with our construction we need to work with a finely representable moduli problem and with the universal abelian scheme associated. To do this, we rigidify the problem adding full level structure.

**Definition 3.3.** Let  $M \geq 3$  be an integer relatively prime to  $N$ . An abelian surface with quaternionic multiplication (QM) by  $\mathcal{R}^{\max}$ , level  $N^+$  structure and full level  $M$  over  $S$  is a quadruple  $(\mathcal{A}, i, G, \nu)$  such that  $(\mathcal{A}, i, G)$  satisfies the properties of definition 3.1 and  $\nu$  is a  $\mathcal{R}^{\max}$ -equivariant isomorphism

$$\nu : (\mathcal{R}^{\max}/M\mathcal{R}^{\max})_S \xrightarrow{\sim} A[M]$$

from the constant group scheme  $(\mathcal{R}^{\max}/M\mathcal{R}^{\max})_S$  to the group scheme of  $M$ -division points.

The corresponding moduli problem admits a fine moduli scheme over  $\mathbb{Q}$  which we denote by  $X_M$ . This is a smooth projective curve over  $\mathbb{Q}$  but it is not geometrically connected. The map induced by forgetting the full level structure

$$q : X_M \rightarrow X$$

is a Galois covering with Galois group  $G/\{\pm 1\}$  with

$$(3.1) \quad G := G_M := (\mathcal{R}^{\max}/M\mathcal{R}^{\max})^\times \cong (\mathcal{R}/M\mathcal{R})^\times \cong GL_2(\mathbb{Z}/M\mathbb{Z}) \cong (R/MR)^\times.$$

Let  $\Gamma_M$  be the group

$$\Gamma_M := \{\gamma \in R^\times : \text{nrd}(\gamma) \equiv 1 \pmod{M}\}$$

**Theorem 3.2** (Čerednik–Drinfeld). *There is an isomorphism of rigid analytic varieties over  $\mathbb{Q}_{p^2}$*

$$(X_{M, \mathbb{Q}_{p^2}})^{\text{an}} \cong \Gamma \setminus (\mathcal{H}_p \times (R/MR)^\times) \cong \coprod_{(\mathbb{Z}/M\mathbb{Z})^\times} \Gamma_M \setminus \mathcal{H}_p.$$

*In particular,  $(X_{M, \mathbb{Q}_{p^2}})^{\text{an}}$  is the disjoint union of Mumford curves  $X_{\Gamma_M} := \Gamma_M \setminus \mathcal{H}_p$ .*

#### 4. MOTIVES ATTACHED TO SHIMURA CURVES

In this section we recall some of the main constructions and properties of motives and Galois representations attached to  $p$ -adic sheaves on Shimura curves.

**4.1. Chow motives and abelian schemes.** In Appendix 10.1 [IS03], Iovita-Spieß generalized the construction of [Sch90] to the case of Shimura curves. We briefly recall some of the main results.

Consider  $\pi : \mathcal{A} \rightarrow X$  the universal abelian scheme associated with the moduli problem of Definition 3.3. The category  $\mathcal{M}^0(X_M)$  of *relative Chow motives* is an additive pseudo-abelian  $\mathbb{Q}$ -linear rigid tensor category. The objects of  $\mathcal{M}^0(X_M)$  are triples  $(Y, p, e)$  where

- $Y$  is a projective  $X_M$ -scheme,
- $p$  is an idempotent element in the ring of relative correspondences  $\text{Corr}_{X_M}(Y \times Y) := \bigoplus_i CH^1(Y_i \times_{X_M} Y)$  where  $Y_i$  are the connected components of  $Y$ ,
- $e$  is an integer.

For a general motive  $\mathcal{M}$  we can define its associated *Chow group*  $CH^i(\mathcal{M})$  to be

$$(4.1) \quad CH^i(\mathcal{M}) := \text{Hom}_{\mathcal{M}^0(X_M)}(\mathbb{1}, \mathcal{M}(i))$$

where  $\mathbb{1}$  is the motive associated to the point. For a smooth projective  $X_M$ -scheme  $Y$  we denote by  $h(Y) = (Y, \Delta_Y, 0)$  its motive. Thanks to an important Theorem of Deninger-Murre [DM91] and Künnemann [Kün94], the motive associated to the abelian scheme  $\mathcal{A}$  decomposes into its graded pieces

$$(4.2) \quad h(\mathcal{A}) = h^0(\mathcal{A}) \oplus h^1(\mathcal{A}) \oplus h^2(\mathcal{A}) \oplus h^3(\mathcal{A}) \oplus h^4(\mathcal{A})$$

such that

$$(4.3) \quad h^i(\mathcal{A}) \cong \wedge^i h^1(\mathcal{A}), \quad h^i(\mathcal{A})^\vee \cong \wedge^i h^{4-i}(\mathcal{A}).$$

The  $\mathcal{R}^{\text{max}}$ -action induces an embedding  $\mathcal{B} \hookrightarrow \text{End}(h^1(\mathcal{A}))$ . Let  $e_2$  be the unique nonzero idempotent in  $\text{End}(h^2(\mathcal{A})) = \text{End}(\wedge^2 h^1(\mathcal{A}))$  such that

$$(4.4) \quad x \cdot e_2 = N(x)e_2 \quad \text{for all } x \in \mathcal{B}$$

where  $N$  is the reduced norm. The projector  $\epsilon_2 \in \text{Corr}_{X_M}(\mathcal{A}, \mathcal{A})$  is defined by

$$(4.5) \quad (\mathcal{A}, \epsilon_2) = h^2(\mathcal{A})_- := \ker(e_2).$$

This motive should be thought as the *non-quaternionic* piece inside  $h^2(\mathcal{A})$ . Let

$$(4.6) \quad (-, -) : \text{Sym}^2(h^2(\mathcal{A})_-) \rightarrow \mathbb{Q}(2)$$

be the restriction of the symmetric pairing

$$(4.7) \quad h^2(\mathcal{A}) \otimes h^2(\mathcal{A}) \cong \wedge^2 h^1(\mathcal{A}) \otimes \wedge^2 h^1(\mathcal{A}) \rightarrow \wedge^4 h^1(\mathcal{A}) \cong \mathbb{Q}(-2).$$

This pairing induce the *Laplace operator*

$$(4.8) \quad \Delta_m : \text{Sym}^m(h^2(\mathcal{A})_-) \rightarrow (\text{Sym}^{m-2}(h^2(\mathcal{A})_-))(-2)$$

given by

$$(4.9) \quad \Delta_m(x_1 x_2 \cdots x_m) = \sum_{1 \leq i < j \leq m} (x_i, x_j) x_1 \cdots \hat{x}_i \cdots \hat{x}_j \cdots x_m.$$

Iovita-Spieß proved that  $\ker(\Delta_m)$  exists as a motive and they denote it as

$$(4.10) \quad (\mathcal{A}^m, \epsilon_{2m}) := \ker(\Delta_m).$$

**Remark 4.1.** In Scholl's original construction [Sch90] for elliptic modular curves  $X$ , the motive of weight  $k$  cusp forms sits directly inside  $\mathrm{Sym}^{k-2}(h^1(\mathcal{E}))$  with  $\mathcal{E} \rightarrow X$  the universal elliptic curve. In this case, the  $h^1(\mathcal{E})$  is already the standard representation of rank 2. On our Shimura curve, parametrising QM abelian surfaces,  $h^1(\mathcal{A})$  is of rank 4 and decomposes as  $V \otimes V'$  where  $V$  is the piece we are interested in. Taking the wedge product gives

$$\wedge^2 h^1(\mathcal{A}) = h^2(\mathcal{A}) = (\mathrm{Sym}^2(V) \otimes \wedge^2 V') \oplus (\wedge^2 V \otimes \mathrm{Sym}^2 V')$$

and consequently the idempotent  $e_2$  isolates the piece  $h^1(\mathcal{A})_- = \mathrm{Sym}^2 V$ . For this reason, the definition of  $(\mathcal{A}^m, \epsilon_n) = \mathrm{Sym}^n V$  is extracted from  $\mathrm{Sym}^m(h^2(\mathcal{A})_-) = \mathrm{Sym}^m(\mathrm{Sym}^2 V)$  through the Laplace operator  $\Delta_m$ .

The Galois group  $G$  defined by (3.1) acts canonically as  $X$ -automorphisms on  $X_M$  and  $\mathcal{A}^m$ . In order to pass from the Shimura curve  $X_M$  to  $X$ , we consider the projector

$$(4.11) \quad p_G := \frac{1}{|G|} \sum_{g \in G} g \in \mathbb{Q}[G]$$

as an element in  $\mathrm{Corr}_X(\mathcal{A}^m, \mathcal{A}^m)$ . The Chow motive  $\mathcal{M}_n$  on  $\mathcal{M}^0(X)$  is then defined to be

$$(4.12) \quad \mathcal{M}_n := (\mathcal{A}^m, p_G \circ \epsilon_n)$$

for  $m = n/2$ .

**4.2.  $p$ -adic local systems.** We follow Sec. 5 [IS03]. In order to determine the  $p$ -adic realization of the motive  $\mathcal{M}_n$ , we consider the  $p$ -adic local system  $R^2\pi_*\mathbb{Q}_p$  and mimic the construction in the light of Remark 4.1. The action of  $\mathcal{R}^{\max}$  on  $\mathcal{A}$  induces an action of  $\mathcal{B}^*$  on  $R^2\pi_*\mathbb{Q}_p$  allowing us to define

$$(4.13) \quad \mathbb{L}_2 := \bigcap_{b \in \mathcal{B}} \ker(b - N(b) : R^2\pi_*\mathbb{Q}_p \rightarrow R^2\pi_*\mathbb{Q}_p)$$

where  $N(b)$  denotes the reduced norm of  $b$ . Consider the Laplace operator

$$(4.14) \quad \Delta_m : \mathrm{Sym}^m \mathbb{L}_2 \rightarrow (\mathrm{Sym}^{m-2} \mathbb{L}_2)(-2)$$

induced by the non-degenerated pairing

$$(4.15) \quad (-, -) : \mathbb{L}_2 \otimes \mathbb{L}_2 \hookrightarrow R^2\pi_*\mathbb{Q}_p \otimes R^2\pi_*\mathbb{Q}_p \rightarrow R^4\pi_*\mathbb{Q}_p \rightarrow \mathbb{Q}_p(-2).$$

For  $n > 2$ , we define  $\mathbb{L}_n$  to be the kernel of  $\Delta_m$  where  $m = n/2$ .

**Lemma 4.1** (Lemma 10.1 [IS03]). *Let  $R_p : \mathcal{M}^0(X_M) \rightarrow D^b(X_M, \mathbb{Q}_p)$  be the  $p$ -adic realization functor from  $\mathcal{M}^0(X_M)$  to the bounded derived category of  $\mathbb{Q}_p$ -sheaves. Let  $m = n/2$ , then we have*

$$R_p(\mathcal{A}^m, \epsilon_n) = \mathbb{L}_n[-n].$$

*If we consider  $\mathcal{M}_n$  as a Chow motive over  $\mathbb{Q}$  under the canonical functor  $\mathcal{M}^0(X) \rightarrow \mathcal{M}^0(\mathbb{Q})$  then its  $p$ -adic realisation is given by*

$$(4.16) \quad R\Gamma(X, R_p(\mathcal{M}_n)) = H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G[-n].$$

The étale cohomology of the  $p$ -adic sheaf  $\mathbb{L}_n$  will be correspond under the  $D_{\mathrm{st}}$  functor to the de Rham cohomology of the  $F$ -isocrystal obtained from the relative de Rham cohomology. In particular, we want to closely study the  $p$ -adic Galois representation

$$(4.17) \quad V := H_{\mathrm{\acute{e}t}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G.$$

**4.3. Cycle map and  $p$ -adic Abel-Jacobi.** In order to compute the  $p$ -adic height pairing of Heegner cycles, we will need to construct the associated extensions of  $p$ -adic Galois representations. To do this, recall that to every element in the Chow group of a Chow motive we can associate its cycle class

$$(4.18) \quad cl : CH^i(\mathcal{M}) \rightarrow R^{2i}\Gamma(X, R_p(\mathcal{M})(i))^{G_K}$$

given by the  $p$ -adic realization of  $\mathrm{Hom}_{\mathcal{M}^0(X_M)}(\mathbb{1}, \mathcal{M}(i))$ . For the motive  $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$  we have

$$(4.19) \quad cl : CH^i(\mathcal{M}) \rightarrow (p_G \circ \epsilon_n)_* H^{2i}(\mathcal{A}^m \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)).$$

We denote the kernel of the cycle class map by  $CH^i(\mathcal{M})_0$ . By Lemma 4.1, the codomain of the cycle map for the motive  $\mathcal{M}_n$  vanishes for every  $i$  and we then have  $CH^i(\mathcal{M}_n) = CH^i(\mathcal{M}_n)_0$ .

The  $p$ -adic Abel-Jacobi map

$$(4.20) \quad cl_0 : CH^i(\mathcal{M})_0 \rightarrow H_{\mathrm{cont}}^1(K, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i))) = \mathrm{Ext}_{\mathrm{Rep}(G_K)}^1(\mathbb{Q}_p, R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)))$$

associates to every class  $z \in CH^i(\mathcal{M})_0$  an extension

$$(4.21) \quad 0 \rightarrow R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) \rightarrow E_z \rightarrow \mathbb{Q}_p \rightarrow 0.$$

The  $p$ -adic representation  $E_z$  is defined as follows. Consider  $\mathcal{M} = (Y, p)$  a relative motive over  $X$ , let  $z \in CH^i(\mathcal{M})_0 = p_* CH^i(Y)_0$  and  $Z$  be a cycle in the Chow group  $CH^i(Y)_0$  such that  $z = p_*[Z]$ . Then we can first consider the Gysin sequence

$$(4.22) \quad 0 \rightarrow H_{\mathrm{et}}^{2i-1}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\mathrm{et}}^{2i-1}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\mathrm{et}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 \rightarrow 0$$

where

$$(4.23) \quad H_{\mathrm{et}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 := \ker(H_{\mathrm{et}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) \rightarrow H_{\mathrm{et}}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)))$$

and then apply the projector  $p$  and take the pullback of the exact sequence by the map  $\mathbb{Q}_p \rightarrow p_* H_{\mathrm{et}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))$  sending 1 to  $cl(z)$

$$(4.24) \quad \begin{array}{ccccccc} 0 & \longrightarrow & R^{2i-1}\Gamma(X, R_p(\mathcal{M})(i)) & \longrightarrow & E_z & \longrightarrow & \mathbb{Q}_p \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow cl \\ 0 & \longrightarrow & p_* H_{\mathrm{et}}^{2i-1}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H_{\mathrm{et}}^{2i-1}((Y - |Z|) \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i)) & \longrightarrow & p_* H_{\mathrm{et}, |Z|}^{2i}(Y \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(i))_0 \longrightarrow 0. \end{array}$$

In the case of  $\mathcal{M}_n = (\mathcal{A}^m, p_G \circ \epsilon_n)$ , using once again the results of Lemma 4.1, we can associate to every  $z \in CH^{m+1}(\mathcal{M})$  an extension

$$(4.25) \quad 0 \rightarrow H_{\mathrm{et}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow E_z \rightarrow \mathbb{Q}_p \rightarrow 0.$$

As discussed in Sec. 7 of [IS03], the extensions obtained for the motive  $\mathcal{M}_n$  are *semistable*, implying that the  $p$ -adic Abel-Jacobi map factors through

$$(4.26) \quad cl_0 : CH^{m+1}(\mathcal{M}_n)_0 \rightarrow H_{\mathrm{st}}^1(K, H_{\mathrm{et}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G)$$

where  $H_{\mathrm{st}}^1$  denotes the Selmer group introduced by Nekovář [Nek93] given by

$$(4.27) \quad H_{\mathrm{st}}^1(K, V) := \ker(H^1(K, V) \rightarrow H^1(K, V \otimes_{\mathbb{Q}_p} B_{\mathrm{st}})).$$

for any  $p$ -adic representation  $V$  of  $G_K$ . If  $V$  is semistable, then  $H_{\mathrm{st}}^1(K, V)$  can be identified with the group of extension classes of  $V$  by  $\mathbb{Q}_p$ . In our case we then have that the  $p$ -adic Abel-Jacobi map factors through

$$(4.28) \quad cl_0 : CM^{m+1}(\mathcal{M}_n) \rightarrow \mathrm{Ext}_{\mathrm{Rep}_{\mathrm{st}}(G_K)}^1(\mathbb{Q}_p, H_{\mathrm{et}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G).$$



**4.4. Biextensions.** Let  $D$  be a reduced effective divisor on  $X_M$  defined over  $K$  such that all closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . There is then an exact sequence of  $G_K$ -modules given by the Gysin sequence

$$(4.29) \quad 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1((X - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x(m) \rightarrow 0.$$

It's important to remark that the direct sum of the stalks of the  $p$ -adic sheaf  $\mathbb{L}_n(m)$  is the  $p$ -adic realisation of the motive

$$(4.30) \quad R\Gamma^{2m}(X_K, R_p((\mathcal{A}_D^m, \epsilon_n)_K)(m)) = \bigoplus_{x \in D} (\mathbb{L}_n)_x(m)$$

where  $\mathcal{A}_D^m = \coprod_{x \in D} \mathcal{A}_x^m$  is the disjoint union of products of abelian surfaces associated with  $x \in D$ .

Its dual exact sequence of  $p$ -adic representations of  $G_K$  is given by the relative cohomology sequence

$$(4.31) \quad 0 \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x(m) \rightarrow H_{\text{ét}, |D|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow 0.$$

The motives  $\mathcal{M}_n$  come equipped with the symmetric pairing (4.6) and at the level of  $p$ -adic realization  $\mathbb{L}$  descends to (4.15) induced by Poincaré duality. In particular, we have

$$(-, -) : \mathbb{L}_n(m) \otimes \mathbb{L}_n(m) \rightarrow \mathbb{Q}_p.$$

The pairing between the exact sequences is then given by

$$(4.32) \quad \begin{array}{ccc} H_{\text{ét}}^1((X - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H_{\text{ét}, |D|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H_{\text{ét}}^2(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G \xrightarrow{\sim} \mathbb{Q}_p \\ \uparrow & \downarrow & \parallel \\ H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \times H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G & \longrightarrow & H_{\text{ét}}^2(X_M \otimes \overline{\mathbb{Q}}, \mathbb{Q}_p(1))^G \xrightarrow{\sim} \mathbb{Q}_p. \end{array}$$

Consider now two effective reduced divisors  $D_1, D_2$  on  $X_M$  defined over  $K_1, K_2$  with disjoint support such that all their closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . Consider the two short exact Gysin sequences

$$(4.33) \quad e_{D_1} : 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m) \rightarrow 0$$

$$(4.34) \quad e_{D_2} : 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1(U_2 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D_2} (\mathbb{L}_n)_x(m) \rightarrow 0$$

is it removing removing a choice of lift or the whole preimage? where  $U_i$  is the open subscheme of  $X_M$  defined by removing a choice of lift of  $D_i$  to  $X_M$ . The cohomology group

$$(4.35) \quad H_{\text{ét}, |D_2|}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G$$

represents a biextension of the exact sequence  $e_{D_1}$  and of the twisted dual  $e_{D_2}^\vee(1)$ . In particular, we obtain a diagram with exact rows and exact columns of  $p$ -adic representations of the absolute Galois group of the composite field  $G_{K_1 K_2}$

$$(4.36) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \bigoplus_{x \in D_2} (\mathbb{L}_n)_x(m+1) & \longrightarrow & H_{\text{ét}, |D_2|}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G & \longrightarrow & H_{\text{ét}}^1(U_1 \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m) & \xlongequal{\quad} & \bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m) \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}.$$

We will construct maps of  $p$ -adic Galois representations  $\mathbb{Q}_p \rightarrow \bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m)$  and  $\bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m+1) \rightarrow \mathbb{Q}_p(1)$  and we will take the pull-back of the first one and the push-out by the second one to obtain the diagram

$$(4.37) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2 & \longrightarrow & V \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \mathbb{Q}_p & = & \mathbb{Q}_p \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array} .$$

**Remark 4.2.** In light of equation (4.30), the problem of constructing trivial subobjects and quotients of  $\bigoplus_{x \in D_1} (\mathbb{L}_n)_x(m)$  will be addressed from the motivic point of view in Section 6 where we will use the construction in [IS03] of suitable classes in  $\mathrm{CH}^m((\mathcal{A}_D^m, \epsilon_n))$  and associated trivial  $p$ -adic representations under the cycle class map (4.18).

This biextension diagram is going to be the starting point of our computations on the  $p$ -adic height pairing of Section 8.

## 5. $F$ -ISOCRYSTALS ON THE $\mathcal{H}_p$

**5.1.  $GL_2 \times GL_2$  representations.** For every representation  $(V, \rho)$  in  $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2 \times GL_2)$  we can construct a convergent filtered  $F$ -isocrystal on the formal model of the  $p$ -adic upper half plane  $\widehat{\mathcal{H}}$ . We denote by  $(V, \rho_1)$  and  $(V, \rho_2)$  the elements of  $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$  obtained restricting  $\rho$  respectively to the first and second component. Conversely, for any  $(V_1, \rho_1), (V_2, \rho_2)$  in  $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$  we can form a  $GL_2 \times GL_2$  representation

$$(V_1, \rho_1) \odot (V_2, \rho_2) = (V_1 \otimes V_2, \rho_1 \otimes 1_{V_2}, 1_{V_1} \otimes \rho_2).$$

For  $V$  in  $\underline{\mathrm{Rep}}_{\mathbb{Q}_p}(GL_2)$  and  $m \in \mathbb{Z}$  we define

$$V\{m\} := V \odot \det^{\otimes m}, \quad \{m\}V := \det^{\otimes m} \odot V.$$

For  $n \geq 0$ , let  $\mathcal{P}_n$  denote the space of all polynomials in  $X, Y$  which are homogeneous of degree  $n$  over  $\mathbb{Q}_p$  with  $GL_2$ -action

$$(5.1) \quad P(X, Y) \cdot \gamma = P(aX + bY, cX + dY), \quad \text{for } \gamma^{-1} = (\det \gamma)^{-1} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Denote the dual representation by  $V_n := \mathcal{P}_n^\vee = \mathrm{Hom}_{\mathbb{Q}_p}(\mathcal{P}_n, \mathbb{Q}_p)$ .

Let  $\widehat{\mathcal{H}}_p$  denote the canonical formal model of  $\mathcal{H}_p$  over  $\mathbb{Z}_p$ . We briefly recall the convergent filtered  $F$ -isocrystals associated to a  $GL_2 \times GL_2$  representation  $(V, \rho_1, \rho_2)$  see [IS03, Section 4] for more details.

- As an *isocrystal*,  $\mathcal{E}(V) := V \otimes_{\mathbb{Q}_p} \mathcal{O}_{\widehat{\mathcal{H}}}$  is given to be the isotrivial vector bundle over the  $p$ -adic upper half plane  $\mathcal{O}_{\widehat{\mathcal{H}}}$  with the induced connection given by usual derivation  $d$ ;
- Consider the endomorphism

$$(5.2) \quad \varphi_V := \rho_2 \begin{pmatrix} 0 & p \\ 1 & 0 \end{pmatrix} : V \rightarrow V.$$

The Frobenius is then given by

$$(5.3) \quad \Phi = \varphi_V \otimes \Phi_{\mathcal{O}_{\mathcal{H}_p}}$$

in particular, it only depends on the  $GL_2$  representation  $\rho_2$ .

- The filtration is only defined in terms of  $\rho_1$ . We describe it only for the representations  $V_n$

$$(5.4) \quad F^k V_n = \sum_{j=0}^{n-k} (zx + y)^k x^j y^{n-j-k} \mathcal{O}_{\mathcal{H}_p}$$

where  $x, y \in V_n$  are the dual elements corresponding to the variables  $X, Y$  in  $\mathcal{P}_n$ . In particular, we have  $F^0 V_n = V_n \otimes_{\mathbb{Q}_p} \mathcal{O}_{\hat{\mathcal{H}}}$ ,  $F^n V_n = (zx + y)^n \mathcal{O}_{\hat{\mathcal{H}}}$  and  $F^{n+1} V_n = 0$ .

In the case where  $\Gamma \subseteq GL_2$  is a discrete, cocompact, and torsion-free subgroup, the filtered isocrystals  $\mathcal{E}(V)$  descend to filtered isocrystals on the Mumford curve  $X_\Gamma$ . In particular, the action of  $GL_2$  on  $\mathcal{E}(V)$  through  $\rho_1$  is compatible with the action on  $\mathcal{H}_p$ . We will denote these filtered isocrystals again by  $\mathcal{E}(V)$  and denote their associated locally free  $\mathcal{O}_{X_\Gamma}$ -module with connection and filtration by  $E(V)$ .

Consider now the  $GL_2 \times GL_2$ -representation  $(M_2, \rho_1, \rho_2)$  given by

$$(5.5) \quad \rho_1(A)(B) := A \cdot B, \quad \rho_2(A)(B) := B \cdot \text{adj}(A)$$

for  $A \in GL_2$  and  $B \in M_2$  where  $\text{adj}(A)$  is the adjugate matrix.

The following result explains why the  $F$ -isocrystals  $\mathcal{E}(V_n\{n/2\})$  will be the correct object of study.

**Theorem 5.1.** *Let  $R_{F\text{-isoc}} : \mathcal{M}_{\text{sm}}^0(X_M) \rightarrow K^b(\text{MF}(X_M))$  be the realization functor from the full subcategory of smooth objects in  $\mathcal{M}^0(X_M)$  to category of bounded complexes of convergent filtered  $F$ -isocrystals given by*

$$(5.6) \quad \begin{aligned} R_{F\text{-isoc}} : \mathcal{M}_{\text{sm}}^0(X_M) &\rightarrow K^b(\text{MF}(X_M)) \\ (Y, p, i) &\mapsto p_* Rf_* \Omega_{Y/X_M}^\bullet[i]. \end{aligned}$$

*I am not sure how to properly write the previous map. What i wrote is defined only for  $Y/X$  smooth. It does not matter for the rest but I wanted to have a notation that could mirror Lemma 3.1 There exists an (up to scaling) canonical isomorphism of filtered isocrystals on  $X_M$*

$$(5.7) \quad R_{F\text{-isoc}}(h^1(\mathcal{A})) \cong \mathcal{E}(M_2, \rho_1, \rho_2)[-1]$$

where  $\mathcal{A}$  is the universal abelian surface over  $X_M$ . This implies that the de Rham cohomology  $\mathcal{H}_{\text{dR}}^1(\mathcal{A}/X_M)$  is isomorphic to the locally free  $\mathcal{O}_{X_M}$ -module  $E(M_2)$  of rank 4.

*Proof.* See [Fal92, Sec. 5] and [IS03, Lemma 5.10]. □

Now we can observe that  $(M_2, \rho_1, \rho_2)$  is canonically isomorphic to  $V_1 \odot V_1$ , giving

$$(5.8) \quad (\wedge^2 M_2, \wedge^2 \rho_1, \wedge^2 \rho_2) = \wedge^2 V_1 \odot \text{Sym}^2 V_1 \oplus \text{Sym}^2 V_1 \odot \wedge^2 V_1 = \{1\} V_2 \oplus V_2 \{1\}.$$

In particular, we obtain

$$(5.9) \quad \mathcal{E}(V_2\{1\}) \cong \bigcap_{b \in \mathcal{B}} \ker(b - N(b) : \mathcal{E}(M_2) \rightarrow \mathcal{E}(M_2))$$

where the action of  $\mathcal{B}$  is given by the isomorphism  $\mathcal{B} \otimes \mathbb{Q}_{p^2} \cong M_2(\mathbb{Q}_{p^2})$ . Observe that  $V_2$  inherits a non-degenerate symmetric bilinear pairing by restriction to

$$(5.10) \quad (-, -) : V_2\{1\} \otimes V_2\{1\} \hookrightarrow \wedge^2 M_2 \otimes \wedge^2 M_2 \rightarrow \wedge^4 M_2 \cong \det^{\otimes 2} \odot \det^{\otimes 2} = \{2\}\{2\}.$$

Compare once again this to the pairings (4.6) and (4.15). We will need an explicit and computable description of this pairing that will be later use to identify Heegner cycles. We recall from [BDIS02, Sec. 1.2] the construction using the following isomorphic description.

**Lemma 5.1.** *Let  $\mathcal{U} = \{U \in M_2 : \text{tr}(U) = 0\}$  be endowed with the right  $\text{GL}_2$  action defined by*

$$(5.11) \quad U \cdot A := \bar{A}UA$$

*for  $U \in \mathcal{U}$  and  $A \in \text{GL}_2$ . There is an isomorphism of right- $\text{GL}_2$ -modules defined by*

$$(5.12) \quad \mathcal{U} \rightarrow \mathcal{P}_2$$

$$U \mapsto P_U(X, Y) := \text{tr} \left( U \begin{pmatrix} X \\ Y \end{pmatrix} (Y \quad -X) \right).$$

The pairing on  $\mathcal{P}_2$  arises from the Cartan-Killing form on  $\mathcal{U}$

$$(5.13) \quad (P_{U_1}, P_{U_2}) := -\text{tr}(U_1 \bar{U}_2)$$

for every  $U_1, U_2 \in \mathcal{U}$ . This induces an isomorphism

$$(5.14) \quad \mathcal{P}_2 \rightarrow V_2$$

$$P(X, Y) \mapsto (P(X, Y), -)$$

and induces a non-degenerate symmetric pairing on  $V_2$  coinciding with (5.10). Consider the Laplace operator

$$(5.15) \quad \Delta_m : \text{Sym}^m V_2 \rightarrow \text{Sym}^{m-2} V_2 \otimes \det^{\otimes 2}$$

defined analogously to (4.8) and (4.14). The  $\text{GL}_2$ -subrepresentation  $V_{2m}$  of  $\text{Sym}^m V_2$  appears without surprise to be the kernel  $\Delta_m$ .

**5.2.  $F$ -isocrystals on  $X_\Gamma$ .** For an even positive integer  $n$  we will denote

$$\mathcal{V}_n := E(V_n\{n/2\})$$

to be the locally free  $\mathcal{O}_{X_\Gamma}$ -module with connection and filtration.

We can now define one of the main cardinal cohomological object.

**Definition 5.1.** *Let  $\Gamma \subseteq \text{GL}_2$  be a discrete and cocompact subgroup. If  $\Gamma$  is torsion-free we define*

$$(5.16) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) := H^1(X_\Gamma, \mathcal{V}_n \otimes \Omega_1^\bullet)$$

*where the complex  $(\mathcal{V}_n \otimes \Omega_1^\bullet)$  is induced by the integrable connection. If  $\Gamma$  is not torsion-free, we consider a torsion-free subgroup  $\Gamma' \subseteq \Gamma$  and we define*

$$(5.17) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) := H_{\text{dR}}^1(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'}$$

In Section 7, we will study the properties of these cohomology groups. In particular, the  $F$ -isocrystal structure of  $\mathcal{E}(V_n\{n/2\})$  will induce the structure of a filtered  $(\varphi, N)$ -module on  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$ . We then have a comparison isomorphism between the  $p$ -adic representations attached to  $\mathbb{L}_n$  and the hypercohomology of the vector bundle with connection  $\mathcal{V}_n$ .

**Important:** I am writing everything base-changing to  $\mathbb{Q}_{p^2}$  and not to  $\mathbb{Q}_p^{ur}$  as IS do. In their paper they do not need to consider precise field extensions and so they can state their results in terms of unramified extensions. I believe we will need to keep track of the finite extensions if we want in the end to say something about in which field extensions our biextension live. This needs to be double checked. I will try to flag it everywhere it will appear.

**Theorem 5.2.** *The  $p$ -adic representation  $H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G$  is semistable and we have a canonical (up to scaling) isomorphisms of filtered  $(\varphi, N)$ -modules*

$$(5.18) \quad D_{\text{st}, \mathbb{Q}_{p^2}}(H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n)^G) = H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n).$$

*Proof.* See [IS03, Theorem 5.9] □

One of the important properties of this theorem and key element inside its proof is the compatibility between the symmetric products defined over the  $p$ -adic local system  $\mathbb{L}_n$  (4.15) and the  $F$ -isocrystal  $\mathcal{E}(V_n\{n/2\})$  (5.10). This has indeed a motivic nature and comes from the pairing (4.6). One instance of this compatibility is the fact that the pairing

$$(5.19) \quad H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n) \otimes H_{\text{ét}}^1(X_M \otimes \bar{\mathbb{Q}}, \mathbb{L}_n) \rightarrow \mathbb{Q}_{p^2}[n+1]$$

is mapped under Fontaine's functor to the pairing

$$(5.20) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \otimes H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \rightarrow H_{\text{dR}}^2(X_\Gamma, \mathcal{V}_n \otimes \mathcal{V}_n) \rightarrow H_{\text{dR}}^2(X_\Gamma, \mathcal{O}_{X_\Gamma})[n] \cong \mathbb{Q}_{p^2}[n+1]$$

where we have used the pairing of isocrystals previously defined

$$(5.21) \quad V_n \otimes V_n \rightarrow \det^{\otimes n}.$$

This property will be used later to pass from dual exact sequences in the category of  $p$ -adic representations to the dual of exact sequences in the de Rham realizations.

Consider now  $D$  to be an effective reduced divisor of  $X$  defined over  $K$ . Assume that  $K/\mathbb{Q}$  is nontrivial unramified at  $p$  and all the closed points  $x_1, x_2, \dots, x_r \in |D|$  are defined over  $K$  and such that all  $x_i$  reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ . We then have a Gysin exact sequence of étale cohomology groups

$$(5.22) \quad e_{\text{ét}} : 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \rightarrow H_{\text{ét}}^1(U \otimes \overline{\mathbb{Q}}, \mathbb{L}_n)^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x(-1) \rightarrow 0.$$

Let  $z_1, z_2, \dots, z_r \in \mathcal{H}_p(K_p)$  points on the  $p$ -adic upper half plane such that  $\Gamma z_i = q(x_i)$  under the  $p$ -adic uniformization of  $X$  Theorem 3.1.

**Theorem 5.3.** *Assume that  $x_1, x_2, \dots, x_r$  specialize to smooth distinct points of the special fiber  $X_{\mathbb{F}_{p^2}}$ . The exact sequence obtained from (5.22) by applying the functor  $D_{\text{st}, K_v}$  is isomorphic to the sequence of filtered  $(\varphi, N)$ -modules*

$$(5.23) \quad e_{\text{dR}} : 0 \rightarrow H_{\text{dR}}^1(X_{\Gamma, K_p}, \mathcal{V}_n) \rightarrow H_{\text{dR}}^1(U_{K_p}, \mathcal{V}_n) \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[1] \rightarrow 0$$

where  $U = X \setminus q(D)$ .

*Proof.* See [IS03, Theorem 5.13]. □

**5.3.  $\mathbb{Q}_{p^2}$ -points on  $\mathcal{H}_p$ .** It will be useful to identify the set of  $\mathbb{Q}_{p^2}$ -points on  $\mathcal{H}_p$  with the set of  $\mathbb{Q}_p$ -algebra homomorphisms  $\text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$ . This will turn out useful when we will describe Heegner points geometrically on  $X$  and  $p$ -adic analytically on  $X_\Gamma$ .

**Lemma 5.2.** *There is a bijection between  $\text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  and  $\mathcal{H}_p(\mathbb{Q}_{p^2})$ .*

*Proof.* Every  $\Psi \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  defines an action of  $\mathbb{Q}_{p^2}^*$  on  $\mathcal{H}_p(\mathbb{Q}_{p^2})$ . The corresponding point  $z_\Psi \in \mathcal{H}_p(\mathbb{Q}_{p^2})$  is defined to be the unique point such that

$$(5.24) \quad \Psi(a) \cdot z = az$$

for every  $a \in \mathbb{Q}_{p^2}^*$ . Conversely, for every point  $z \in \mathcal{H}(\mathbb{Q}_{p^2})$  we can define the associated homomorphism  $\Psi_z \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  to be the linear transformation associated to multiplication by  $\mathbb{Q}_{p^2}^*$  on the projective coordinates of  $z$ . These two maps can be easily checked to be inverse of each other. □

Consider now a point  $z \in \mathcal{H}_p(\mathbb{Q}_{p^2})$  and let  $\Psi_z \in \text{Hom}(\mathbb{Q}_{p^2}, M_2(\mathbb{Q}_p))$  we want to compute the stalk of the isocrystal  $\mathcal{E}(V_n\{n/2\})_{\mathbb{Q}_{p^2}}$  at the point  $z$ . This in particular will have the structure of a *filtered Frobenius module*. As a vector space this coincide with  $V_{n, \mathbb{Q}_{p^2}} = V_n \otimes \mathbb{Q}_{p^2}$  and it is generated by  $\text{ev}_z \in V_{n, \mathbb{Q}_{p^2}} = \text{Hom}(\mathcal{P}_n, \mathbb{Q}_{p^2})$  given by

$$(5.25) \quad \text{ev}_z(P(X, Y)) := P(z, 1)$$

that has the property

$$(5.26) \quad (\Psi_z(a) \cdot \text{ev}_z)(P(X, Y)) = a^n P(z, 1).$$

To define the filtration, consider first the subspaces

$$(5.27) \quad (V_{n, \mathbb{Q}_{p^2}})_i := \{v \in V_{n, \mathbb{Q}_{p^2}} : \rho_1(\Psi(a))(v) = a^i \sigma(a)^{n-i} v \quad \forall a \in \mathbb{Q}_{p^2}^*\}$$

and then define

$$(5.28) \quad F^i(V_{n, \mathbb{Q}_{p^2}})_z := \bigoplus_{j \geq i} (V_{n, \mathbb{Q}_{p^2}})_i.$$

The Frobenius endomorphism is given by  $\phi_V \otimes \sigma$ .

## 6. HEEGNER CYCLES

In this section we will discuss special cycles in  $\mathrm{CH}^{m+1}(\mathcal{M}_n)$  arising from abelian surfaces with complex multiplication. Under the Čerednik-Drinfelĭ  $p$ -adic uniformization, we will also study their characterization on the  $p$ -adic upper half plane. We will follow the construction of [IS03] Sec.8.

Recall that we want to construct interesting classes in  $\mathrm{CM}^{m+1}(\mathcal{M}_n)$  or, as explained at the end of Section 4, construct trivial  $p$ -adic representations in the  $p$ -adic realization of motives  $(\mathcal{A}_D^m, \varepsilon_n)$  for certain reduced effective divisors  $D$  on  $X_M$ . We will define these classes using special CM abelian surfaces.

### 6.1. Heegner points and special cycles.

**Definition 6.1.** Consider  $A$  an abelian surface with QM by  $\mathcal{R}^{\max}$ . The abelian surface is said to have complex multiplication (CM) if  $\mathrm{End}_{\mathcal{R}^{\max}}(A) \neq 0$ . In this case it is an order  $\mathcal{O}$  in an imaginary quadratic number field. A closed point on  $X$  is called CM if it can be represented by a triple  $(A, \iota, C)$  where  $A$  has complex multiplication. It is furthermore called Heegner point if  $(A, \iota, C)$  is CM by an order  $\mathcal{O}$  and  $C$  is  $\mathcal{O}$ -stable.

We will now fix a number field  $K$  satisfying certain prescribed properties that will be key in the construction of the Chow classes.

**Hypothesis 6.1** (Heegner). Let  $K$  be an imaginary quadratic field of discriminant  $D_K$  satisfying the following properties:

- $(D_K, N) = 1$ ,
- all prime divisors of  $N^- p$  are inert in  $K$ ,
- all prime divisors of  $N^+$  are split in  $K$ .

Consider now an abelian surface  $A$  with CM by the maximal order  $\mathcal{O}_K$ . Thanks to the splitting behaviour of the primes in  $K$  we obtain

$$\mathrm{End}(A) \cong \mathcal{O}_K \otimes \mathcal{R}^{\max} \cong M_2(\mathcal{O}_K).$$

In particular, fixing an isomorphism  $\mathrm{End}(A) \cong M_2(\mathcal{O}_K)$ , we get an isomorphism

$$(6.1) \quad A \cong E \times E$$

where  $E$  is an elliptic curve with CM over  $H$  Hilbert class field of  $K$  and  $\mathrm{End}(E) \cong \mathcal{O}_K$ . As usual, we fix an isomorphism between the endomorphism ring and the maximal order. The special rigid structure of  $A$  is reflected in its associated motive. By the Künneth decomposition we have

$$(6.2) \quad \begin{aligned} h^2(A) &= h^2(E \times E) = (h^2(E) \otimes h^0(E)) \oplus (h^1(E) \otimes h^1(E)) \oplus (h^0(E) \otimes h^2(E)) = \\ &= \mathbb{1}(-1) \oplus (\mathrm{Sym}^2(h^1(E)) \oplus \wedge^2 h^1(E)) \oplus \mathbb{1}(-1) = \\ &= \mathbb{1}(-1)^{\oplus 3} \oplus \mathrm{Sym}^2(h^1(E)) \end{aligned}$$

One comment here, feel free to ignore but I found the other decomposition that you wrote before to be more illuminating.  $E$  has CM by  $K$ , and  $A$  QM by  $B$  and therefore  $B \otimes_{\mathbb{Q}} K$  is identified with  $M_2(K) = K \otimes M_2(\mathbb{Q})$ . Then the action of  $K$  and  $B$  can be described on

$$h^1(A) = h^1(E \times E) = h^1(E) \otimes_{\mathbb{Q}} \mathbb{Q}^2,$$

where  $K$  acts on the first factor and  $M_2(K)$  acts on the second

$$h^2(A) = \wedge^2 h^1(A) = \wedge^2(h^1(E) \otimes_{\mathbb{Q}_p} \mathbb{Q}_p^2) = (\wedge^2 h^1(E) \otimes \mathrm{sym}^2(\mathbb{Q}_p^2)) \oplus (\mathrm{sym}^2 h^1(E) \otimes \wedge^2 \mathbb{Q}_p^2)$$

As a decomposition of  $K$ -modules this is the same decomposition as you wrote, but I like it because it also shows the  $B$ -action. Namely it is clear that on the  $\mathrm{Sym}^2$  part  $M_2$  acts by determinant

and therefore  $B$  acts by the reduced norm while on the other part it acts irreducibly on a three dimensional vector space. and it can be shown that the action of  $\mathcal{B}^*$  cuts, in the notation of (4.5), the part

$$(6.3) \quad h^2(A)_- = \text{Sym}^2(h^1(E)).$$

I really like your presentation! We can discuss it next time and I will then implement it. Furthermore, since  $E$  has CM by  $\mathcal{O}_K$ , then  $h^1(E)$  acquires an action of  $K$  and in particular the base change to  $K$  will decompose into two summands

$$(6.4) \quad h^1(E) \otimes_{\mathbb{Q}} K = M(\psi) \oplus M(\bar{\psi})$$

where  $M(\psi), M(\bar{\psi})$  are the motives associated to the Hecke character  $\psi$  attached to field extension  $K/\mathbb{Q}$  and its conjugate as in Ch. 1 [Sch88]. From the fact that  $\psi\bar{\psi} = N_{K/\mathbb{Q}}$ , the symmetric square is then given by

$$(6.5) \quad \text{Sym}^2(h^1(E)) \otimes_{\mathbb{Q}} K = M(\psi^2) \oplus \mathbb{1}(-1) \oplus M(\bar{\psi}^2).$$

Consider the cycle defined by

$$(6.6) \quad y := [\Gamma_{\sqrt{D_K}}] + D_K[\{O\} \otimes E] - [\{O\} \otimes E] \in CH^1(E \times E)$$

where  $\Gamma_{\sqrt{D_K}}$  is the graph of the endomorphism  $\sqrt{D_K}$  in  $\text{End}(E)$ . Sec. 8 [IS03] proves that  $y \in CH^1(A, \epsilon_2)$  and will land in the  $\mathbb{1}$ -part of (6.5) under the cycle class map.

Let  $t$  be the number of prime factors of  $N$  and  $h$  the class number of  $K$ . Consider the following subgroups of  $\text{Aut}(X_H)$ :

- $\mathcal{W} \cong (\mathbb{Z}/2\mathbb{Z})^t$  subgroup generated by *Atkin-Lehner involutions*  $\omega_l$  for every  $l$  prime divisor of  $N$ ,
- $\Delta = \text{Pic}(\mathcal{O}_K)$  subgroup identified with  $\text{Gal}(H/K)$  via the Artin map.

By Lemma 2.5 [BD96] there are precisely  $2^t h$  Heegner points with CM by  $\mathcal{O}_K$  which are defined over the Hilber class field  $H$  of  $K$  and the group action of  $\mathcal{W} \times \Delta$  on this set is free and transitive.

In order to construct the desired cycle classes in  $CH^{m+1}(\mathcal{M}_n)$  we need once again to pass through the fine moduli scheme  $X_M$ . Consider  $P$  closed Heegner point on  $X_H$  and let  $\tilde{P}$  a closed point in the fiber  $q^{-1}(P)$  in  $(X_M)_H$ . The fiber  $\mathcal{A}_{\tilde{P}}$  above  $\tilde{P}$  is an abelian surface with CM by  $\mathcal{O}_K$ . We can then consider  $y_{\tilde{P}} \in CH^1(\mathcal{A}_{\tilde{P}}, \epsilon_2)$  as defined above. For an abelian surface with CM, the cycle class map factors through algebraic equivalence, in particular we could modify the cycle  $y_{\tilde{P}}$  by any element in  $\text{Pic}^0(\mathcal{A}_{\tilde{P}})$ . We choose the representatives

$$(6.7) \quad z_{\tilde{P}} \in [y_{\tilde{P}}] \in \text{NS}(\mathcal{A}_{\tilde{P}}) = \text{Pic}(\mathcal{A}_{\tilde{P}})/\text{Pic}^0(\mathcal{A}_{\tilde{P}})$$

for every  $\tilde{P}$  in such a way that they are compatible with the  $G$ -action, i.e.

$$(6.8) \quad g_*(z_{\tilde{P}}) = z_{g(\tilde{P})}.$$

**Definition 6.2.** Define the cycle class  $y_P = y_P^{(2)} \in CH^2((\mathcal{M}_n)_H)$  to be the image of  $z_{\tilde{P}}$  under

$$(6.9) \quad CH^1(\mathcal{A}_{\tilde{P}}, \epsilon_2) \xrightarrow{(p_{G \circ \iota_{\tilde{P}}})_*} CH^2((\mathcal{M}_2)_H)$$

where  $\iota_{\tilde{P}} : \mathcal{A}_{\tilde{P}} \hookrightarrow \mathcal{A}_H$ . For arbitrary even  $n \geq 2$ , first consider the  $m$ -th product of cycles under the projector  $\epsilon_n$  as defined in (4.10)

$$(6.10) \quad z_{\tilde{P}}^{(n)} = (\epsilon_n)_*(z_{\tilde{P}} \times \cdots \times z_{\tilde{P}})$$

and then we define the cycle class  $y_P^{(n)} \in CH^{m+1}((\mathcal{M}_n)_H)$  to be the image of  $z_{\tilde{P}}^{(n)}$  under

$$(6.11) \quad CH^m(\mathcal{A}_{\tilde{P}}^m, \epsilon_n) \xrightarrow{(p_{G \circ \iota_{\tilde{P}}})_*} CH^{m+1}((\mathcal{M}_n)_H).$$

Thanks to the compatibility under the  $G$ -action (6.8), the definition of  $y_P$  and  $y_P^{(n)}$  is independent of the point  $\tilde{P}$ , but still depends on the set of compatible classes  $z_{\tilde{P}}$ .

**6.2. Heegner points on  $\mathcal{H}_p$ .** We briefly recall the description of  $p$ -adic analytic Heegner points on  $\mathcal{H}_p$  following [BD98] and [IS03].

**Definition 6.3.** An embedding  $\Psi : K \rightarrow B$  is called optimal if  $\Psi(\mathcal{O}_K[1/p]) \subseteq R$ . The group  $\Gamma$  acts by conjugation of the set of optimal embeddings and we denote  $\text{emb}(\mathcal{O}_K, R)$  the set of conjugacy classes  $[\Psi]$ .

The set  $\text{emb}(\mathcal{O}_K, R)$  is endowed with an action of  $\Delta$  and [BD98, Theorem 5.3] gives a bijection between the set of Heegner points on  $X_H$  and  $\text{emb}(\mathcal{O}_K, R)$  compatible with the  $\Delta$ -action. Fix an embedding  $H \hookrightarrow \mathbb{C}_p$  and let  $v|p$  the corresponding place of  $H$ . Since  $p$  is inert in  $K$  and totally split in  $H$ , we then identify  $K_p$  and  $H_v$  with  $\mathbb{Q}_{p^2}$ . By Lemma 5.2, we can identify the classes in  $\text{emb}(\mathcal{O}_K, R)$  with  $K_p$ -points of  $X$  via

$$(6.12) \quad \text{emb}(\mathcal{O}_K, R) \hookrightarrow \Gamma \backslash \text{Hom}(K_p, M_2(\mathbb{Q}_p)) = X_\Gamma(K_p).$$

An Heegner point  $P \in X_\Gamma(K_p)$  lies in the image of  $\text{emb}(\mathcal{O}_K, R)$  since the corresponding embedding  $\Psi_P$  can be defined by looking at the endomorphism ring of the associate CM abelian surface and at its reduction modulo  $p$

$$(6.13) \quad \Psi_P : \mathcal{O}_K = \text{End}(\mathcal{A}_P) \hookrightarrow \text{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2}).$$

By the work of Čerednik–Drinfeld [BC91], we have  $\text{End}(\mathcal{A}_P \otimes \mathbb{F}_{p^2})[1/p] = R[1/p]$ . This map is an optimal embedding by [GZ86, Prop. 7.3]. See [BD98] for the complete details.

## 7. DE RHAM COHOMOLOGY AND BRUHAT-TITS COCYCLES

In order to compute the local  $p$ -adic component of Nekovar height, we need to understand the filtered  $(\phi, N)$ -structure of the following cohomology groups

$$H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n), \quad H_{\text{dR}}^1(U, \mathcal{V}_n), \quad H_{\text{dR}}^1(X_\Gamma, Z, \mathcal{V}_n), \quad H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$$

where  $U \subseteq X_\Gamma$  is an open set and  $Z \hookrightarrow X_\Gamma$  is a closed subscheme belonging to  $U$ . When  $\Gamma$  contains torsion, these groups are defined as follows: Pick a normal subgroup  $\Gamma' \subset \Gamma$  which is torsion-free. Let  $\pi : X_{\Gamma'} \rightarrow X_\Gamma$  be the projection,  $U' = \pi^{-1}(U)$  and  $Z' = \pi^{-1}(Z)$ , then define

$$\begin{aligned} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) &:= H_{\text{dR}}^1(X_{\Gamma'}, \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(U, \mathcal{V}_n) &:= H_{\text{dR}}^1(U', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(X_\Gamma, Z, \mathcal{V}_n) &:= H_{\text{dR}}^1(X_{\Gamma'}, Z', \mathcal{V}_n)^{\Gamma/\Gamma'} \\ H_{\text{dR}}^1(U, Z, \mathcal{V}_n) &:= H_{\text{dR}}^1(U', Z', \mathcal{V}_n)^{\Gamma/\Gamma'}. \end{aligned}$$

These groups do not depend on the choice of  $\Gamma'$ .

Here I guess what we actually want is I think the cohomology of  $X_\Gamma$  and not that of  $X$ . For the remainder of the paper, fix a choice of branch of  $p$ -adic logarithm  $\log : \mathbb{C}_p^\times \rightarrow \mathbb{C}_p$ .

**7.1. The  $(\phi, N)$ -module structure of  $H_{\text{dR}}^1$ .** In this section we recall the structure of  $H_{\text{dR}}^1(X, \mathcal{V}_n)$  as a  $(\phi, N)$ -module following [IS03] and [Mas11]. Assume for now that  $\Gamma$  is torsion-free. **check the following:** Then the Mumford curve  $X_\Gamma$  over  $\mathbb{Q}_p^{\text{ur}}$  is a smooth projective curve with a semistable model  $\mathfrak{X}_\Gamma$ . Let  $C_1, \dots, C_r$  be its irreducible components of the special fiber,  $C$ , of  $\mathfrak{X}_\Gamma$ . Let  $\mathfrak{G} = (\mathfrak{V}(\mathfrak{G}), \mathfrak{E}(\mathfrak{G}))$  the oriented intersection graph of  $C$ . Consider the reduction map

$$\text{red} : X_\Gamma^{\text{an}} \rightarrow C(\overline{\mathbb{F}}_p)$$

and define the admissible cover  $\{U_v\}_v \in \mathfrak{V}$  where  $U_v = \text{red}^{-1}(C_v)$ . For every edge  $e \in \mathfrak{E}$  denote  $o(e)$  the origin vertex and  $t(e)$  the target vertex. We can associate to each edge  $e$  a wide open annulus  $A_e$  given by

$$A_e = U_{o(e)} \cap U_{t(e)} = \text{red}^{-1}(x)$$

where  $x = C_{o(e)} \cap C_{t(e)}$ .



**7.2. The general case.** Let  $D = (\tilde{z}_1) + \cdots + (\tilde{z}_r)$  and  $Z = (\tilde{w}_1) + \cdots + (\tilde{w}_s)$  be disjoint divisors in  $X_\Gamma$  such that each of the points  $\tilde{z}_i$  and  $\tilde{w}_j$  reduces to a smooth point in  $\mathfrak{X}_\Gamma$ . We will denote by  $U$  the open subscheme  $X_\Gamma - D$ , and by

$$i : Z \hookrightarrow X_\Gamma$$

the closed subscheme associated with the same divisor. Our aim is to describe the  $(\varphi, N)$ -module structure on the relative de Rham cohomology group  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$ .

Recall that  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is defined as follows: Let  $\Omega_{(U,Z)}^\bullet(\mathcal{V}_n)$  denote the cone associated to the morphism of complexes

$$i^* \Omega_U^\bullet \otimes \mathcal{V}_n \rightarrow i_*(\Omega_Z^\bullet \otimes i^* \mathcal{V}_n).$$

Then  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is the hypercohomology of the complex of sheaves  $\Omega_{(U,Z)}^\bullet(\mathcal{V}_n)$ .

For each  $v \in \mathfrak{V}$ , let  $U_{v,D}$  denote the wide-open affinoid  $U_v \setminus |D|$ . Then the collection  $(U_{v,D})_{v \in \mathfrak{V}}$  is an admissible covering of  $U$ . Computing  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  by Čech hypercohomology with this choice of covering, each cohomology class is represented by a cocycle

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \quad \text{for } \omega_v \in H^0(U_{v,D}, \Omega_{U_{v,D}}^1 \otimes \mathcal{V}_n^{an}), \alpha \in H^0(Z, i^* \mathcal{V}_n), f_e \in H^0(A_e, \mathcal{V}_n),$$

which satisfy

$$\omega_{v_1}|_{A_e} - \omega_{v_2}|_{A_e} = df_e \quad \text{for every } e = (v_1, v_2) \in \mathfrak{E}.$$

A representative cocycle for a cohomology class is well-defined up to

$$(7.1) \quad ((d\omega_v)_{v \in \mathfrak{V}}, (g_v)|_Z, (g_{s(e)} - g_{t(e)})_{e \in \mathfrak{E}}) \quad \text{for } g_v \in H^0(U_{v,D}, \mathcal{V}_n^{an}).$$

This implies that the map

$$\gamma_{U,Z} : H_{\text{dR}}^1(U, Z, \mathcal{V}_n) \longrightarrow \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1)$$

given by

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \mapsto (\omega_v)_{v \in \mathfrak{V}}$$

is well-defined and surjective, where

$$H_{\mathfrak{V},D}^i = \bigoplus_{v \in \mathfrak{V}} H_{\text{dR}}^i(U_{v,D}, \mathcal{V}_n^{an}), \quad H_{\mathfrak{E}}^i = \bigoplus_{e \in \mathfrak{E}} H_{\text{dR}}^i(A_e, \mathcal{V}_n^{an}).$$

The group  $\ker(\gamma_{U,Z})$  is generated by elements of the form

$$((0)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \quad \text{such that } df_e = 0,$$

modulo elements of the form in (7.1), under the condition that  $d\omega_v = 0$  for all  $v \in \mathfrak{V}$ . Hence the kernel of  $\gamma_{U,Z}$  is naturally identified with the quotient

$$\left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0.$$

We will denote this identification by  $\iota_{U,Z}$  and note that we have an exact sequence

$$(7.2) \quad 0 \rightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0 \xrightarrow{\iota_{U,Z}} H_{\text{dR}}^1(U, V, \mathcal{V}_n) \xrightarrow{\gamma_{U,Z}} \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1) \rightarrow 0.$$

Consider the retraction map

$$P_{U,Z} : H_{\text{dR}}^1(U, V, \mathcal{V}_n) \longrightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{V},D}^0$$

defined on representatives by

$$((\omega_v)_{v \in \mathfrak{V}}, \alpha, (f_e)_{e \in \mathfrak{E}}) \mapsto ((f_e - (F_{o(e)} - F_{t(e)}))_{e \in \mathfrak{E}}, \alpha - (F_v)|_Z),$$

where  $F_v$  is a choice of Coleman primitive of  $\omega_v$  on  $U_{v,D}$  with respect to the fixed choice of  $p$ -adic logarithm, which is well-defined up to constant sections of  $\mathcal{V}_n|_{U_{v,D}}$ . This defines a section

$$T_{U,Z} : \ker(H_{\mathfrak{V},D}^1 \rightarrow H_{\mathfrak{E}}^1) \longrightarrow H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$$

by

$$\alpha \mapsto \beta - \iota_{U,Z}(P_{U,Z}(\beta)),$$

where  $\beta \in H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is any element satisfying  $\gamma_{U,Z}(\beta) = \alpha$ . Denote by

$$\varphi_1 : \ker(H_{\mathfrak{A},D}^1 \rightarrow H_{\mathfrak{E}}^1) \longrightarrow \ker(H_{\mathfrak{A},D}^1 \rightarrow H_{\mathfrak{E}}^1)$$

and

$$\varphi_2 : \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{A},D}^0 \longrightarrow \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{A},D}^0$$

the respective natural Frobenius action. The Frobenius action on  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  is defined by

$$\varphi(\omega) = \iota_{U,Z}(\varphi_2(P_{U,Z}(\omega))) + T_{U,Z}(\varphi_1(\gamma_{U,Z}(\omega))) \quad \text{for } \omega \in H_{\text{dR}}^1(U, Z, \mathcal{V}_n).$$

Given  $e \in \mathfrak{E}$ , there is a natural residue map

$$\text{Res}_e : H_{\text{dR}}^1(A_e, \mathcal{V}_n^{an}) \longrightarrow H_{\text{dR}}^0(A_e, \mathcal{V}_n^{an}).$$

The monodromy operator is defined as

$$N(\omega) := \iota_{U,Z}((\text{Res}_e(\omega|_{A_e}))_{e \in \mathfrak{E}}, 0) \quad \text{for } \omega \in H_{\text{dR}}^1(U, Z, \mathcal{V}_n).$$

**Proposition 7.1.** *The admissible filtered  $(\varphi, N)$ -module  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  admits a slope decomposition*

$$(7.3) \quad H_{\text{dR}}^1(U, Z, \mathcal{V}_n) = H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_m \oplus H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_{m+1}$$

and  $\ker(N) = \text{im}(\iota_{U,Z}) = H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$ ,  $\text{im}(T_{U,Z}) = \ker(P) = H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$ . Furthermore, there is a decomposition

$$(7.4) \quad H_{\text{dR}}^1(U, Z, \mathcal{V}_n) = F_{U,Z}^{m+1} \oplus H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_m.$$

Here  $F_{U,Z}^{m+1} = H^0(U, \Omega_U \otimes \mathcal{V}_n)$ .

**7.3. Pulling back to  $\mathcal{H}_p$ .** By pulling back to  $\mathcal{H}_p$ , the decomposition of  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  obtains a more explicit combinatorial form, which allows us to define a  $(\varphi, N)$ -module for  $\Gamma$  containing torsion. For now, assume as before that  $\Gamma$  is torsion free.

Let  $\mathcal{H}_p := \mathbb{P}^1(\mathbb{C}_p) \setminus \mathbb{P}^1(\mathbb{Q}_p)$  denote the  $p$ -adic upper half-plane. It is a rigid analytic space with an action of  $\text{GL}_2(\mathbb{Q}_p)$  by Mobius transformation.

Denote by  $\mathcal{T}_p = (\mathcal{T}_p^0, \mathcal{T}_p^1)$  the Bruhat-Tits tree with vertex and oriented edge set  $\mathcal{T}_p^0$  and  $\mathcal{T}_p^1$ , respectively. It is equipped with the usual action of  $\text{GL}_2(\mathbb{Q}_p)$ , and there is a  $\text{GL}_2(\mathbb{Q}_p)$ -equivariant reduction map

$$\text{red}_p : \mathcal{H}_p \longrightarrow \mathcal{T}_p.$$

Let  $C^0(V_n)$  (resp.  $C^1(V_n)$ ) denote the  $\text{GL}_2(\mathbb{Q}_p)$  module of maps  $f : \mathcal{T}_p^0 \rightarrow V_n$  (resp.  $f : \mathcal{T}_p^1 \rightarrow V_n$  such that  $f(e) = -f(\bar{e})$  for every oriented edge  $e \in \mathcal{T}_p^1$ ). Define the map

$$\Delta : C^0(V_n) \longrightarrow C^1(V_n)$$

by

$$\Delta(f)(e) = f(s(e)) - f(t(v)) \quad \text{for } e \in \mathcal{T}_p^1,$$

where  $s(e)$  and  $t(v)$ , respectively, are the start and end vertices of  $e$ . This map sits in a short exact sequence

$$(7.5) \quad 0 \longrightarrow V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0.$$

Let  $\nabla : C^1(V_n) \longrightarrow C^0(V_n)$  be the  $\text{GL}_2(\mathbb{Q}_p)$ -equivariant map obtained by

$$\nabla(f)(v) = \sum_{s(e)=v} f(e) \quad \text{for } f \in C^1(V_n), v \in \mathcal{T}_p^0,$$

where the sum runs over all edges starting at the vertex  $v$ . Let  $C_{\text{har}}^1$  denote the kernel of  $\nabla$ . It sits in the exact sequence

$$0 \longrightarrow C_{\text{har}}^1(V_n) \longrightarrow C^1(V_n) \longrightarrow C^0(V_n) \longrightarrow 0.$$

The oriented intersection graph,  $\mathfrak{G}$ , is canonically identified with  $\Gamma \backslash \mathcal{T}_p$ . For each vertex  $v \in \mathcal{T}_p^0$ , let  $N(v)$  be the neighborhood of  $v$  in  $\mathcal{T}_p$ . The reduction maps and the quotient by the action of  $\Gamma$  fit into the commutative diagram

$$\begin{array}{ccc} \mathcal{H}_p & \xrightarrow{\text{red}_p} & \mathcal{T}_p \\ \downarrow & & \downarrow \\ X_\Gamma^{\text{an}} & \xrightarrow{\text{red}} & \mathfrak{G}. \end{array}$$

It follows that for each vertex  $v \in \mathfrak{V}(\mathfrak{G})$ , the wide open affinoid domain  $U_v$  is isomorphic to  $\mathbb{P}^1$  minus a set of disjoint disks, and for each edge  $e \in \mathfrak{E}(\mathfrak{G})$ , the preimage  $A_e$  is indeed a wide open annulus. Therefore, there are compatible isomorphisms

$$C^0(V_n)^\Gamma \cong H_{\mathfrak{G}}^0$$

and

$$C^1(V_n)^\Gamma \cong (H_{\mathfrak{E}}^0)^-.$$

Let  $w_i \in \mathcal{H}_p$ , be any choice of left of  $\tilde{w}_i \in X_\Gamma(\mathbb{C}_p) \cong \mathcal{H}_p/\Gamma$ . It follows that there is an isomorphism

$$\text{Func}\left(\bigcup_{i=1}^s \Gamma \cdot w_i, V_n\right)^\Gamma \cong H^0(Z, i^* \mathcal{V}_n).$$

Note then that  $((H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n)) / H_{\mathfrak{G}, D}^0$  is identified with the first cohomology of the total complex of the bi-complex

$$(7.6) \quad \begin{array}{ccc} C^1(V_n)^\Gamma & \longrightarrow & 0 \\ \uparrow & & \uparrow \\ C^0(V_n)^\Gamma & \longrightarrow & \text{Func}(\bigcup_i \Gamma \cdot w_i, V_n)^\Gamma. \end{array}$$

Since the complex

$$V_n \longrightarrow C^0(V_n) \longrightarrow C^1(V_n) \longrightarrow 0$$

is a free resolution of  $V_n$ , the cohomology of (7.6) computes the  $\Gamma$ -hypercohomology of the complex

$$\mathcal{K}_Z(V_n) = \left( V_n \longrightarrow \text{Func}\left(\bigcup_i \Gamma \cdot w_i, V_n\right) \right),$$

where the arrow indicates the map

$$\alpha \mapsto (w \mapsto \alpha).$$

there is an isomorphism

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong \left( (H_{\mathfrak{E}}^0)^- \oplus H^0(Z, i^* \mathcal{V}_n) \right) / H_{\mathfrak{G}, D}^0.$$

**Definition 7.1.** Let  $\{z_1, \dots, z_r\} \subset \mathcal{H}_p$  be a fixed choice of lifts of the points  $\tilde{z}_i$  to  $\mathcal{H}_p$ , and write

$$\mathcal{H}_U := \mathcal{H}_p - \bigcup_{i=1}^r \Gamma \cdot z_i.$$

Let  $\mathcal{T}_U$  be the tree whose set of vertices is  $\mathcal{T}_p^0 \cup \bigcup_{i=1}^r \Gamma \cdot z_i$ , and whose set of edges is the set of edges of  $\mathcal{T}_p$  along with an edge  $e_w$  for each  $w \in \bigcup_{i=1}^r \Gamma \cdot z_i$ , where the edge  $e_w$  connects the vertex associated with  $w$  and the vertex  $\text{red}_p(w)$ .

Let  $\mathcal{T}_U^1$  denote the set of oriented edges of  $\mathcal{T}_U$ , and  $C_U^1(V_n)$  be the set of functions

$$c : \mathcal{T}_U^1 \longrightarrow V_n$$

satisfying

$$c(\bar{e}) = -c(e) \quad \text{for all } e \in \mathcal{T}_U^1,$$

and

$$\sum_{s(e)=v} c(e) = 0 \quad \text{for all } v \in \mathcal{T}_p^1.$$

**Proposition 7.2.** *There is a natural identification*

$$C_U^1(V_n)^\Gamma \cong \ker(H_{\mathfrak{B},D}^1 \rightarrow H_{\mathfrak{E},D}^1).$$

*Proof.* The group  $\ker(H_{\mathfrak{B},D}^1 \rightarrow H_{\mathfrak{E},D}^1)$  is isomorphic to

$$(7.7) \quad \ker \left( \left( \bigoplus_{v \in \mathcal{T}_p^0} H_{\text{dR}}^1((\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v)), \mathcal{V}_n^{an}) \right)^\Gamma \rightarrow \left( \bigoplus_{e \in \mathcal{T}_p^1} H_{\text{dR}}^1(\text{red}_p^{-1}(e), \mathcal{V}_n^{an}) \right)^\Gamma \right).$$

For every  $v \in \mathcal{T}_p^0$ , the pre-image

$$(\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v))$$

is  $\mathbb{P}^1(\mathbb{C}_p)$  minus a set of  $(p+1)$  disjoint disks and minus a finite set of points. Each disk corresponds to an edge in  $N(v)$  and each removed point corresponds to some  $w \in \bigcup_{i=1}^r \Gamma \cdot z_i$  with  $\text{red}_p(w) = v$ . It follows that an element of the de Rham cohomology group

$$H_{\text{dR}}^1((\text{red}_p|_{\mathcal{H}_U})^{-1}(N(v)), \mathcal{V}_n^{an})$$

is determined by the residues on the removed disks and on the removed points, with the restriction that the sum of residues is 0.  $\square$

It follows that we have an exact sequence

$$(7.8) \quad 0 \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H_{\text{dR}}^1(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C_U^1(V_n)^\Gamma \rightarrow 0$$

The natural restriction map

$$C_U^1(V_n)^\Gamma \longrightarrow C^1(V_n)^\Gamma$$

defines a map

$$\text{res} : C_U^1(V_n) \longrightarrow H^1(\Gamma, \mathcal{K}_Z(V_n))$$

and the monodromy operator can be described as the composition

$$N : H_{\text{dR}}^1(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C_U^1(V_n)^\Gamma \xrightarrow{\text{res}} H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H_{\text{dR}}^1(U, Z, \mathcal{V}_n).$$

**7.4. Removing the restriction on  $\Gamma$ .** If  $\Gamma$  is not necessarily torsion free, choose some normal subgroup  $\Gamma' \subset \Gamma$ , which is torsion free and let  $G = \Gamma/\Gamma'$ . Let

$$r : X_{\Gamma'} \longrightarrow X_\Gamma$$

be the natural reduction map. Let  $D' = r^*D$ ,  $Z' = r^*Z$ , and  $U' = r^{-1}(U)$ . We define

$$H_{\text{dR}}^1(U, Z, \mathcal{V}_n) := H_{\text{dR}}^1(U', Z', \mathcal{V}_n)^G.$$

The group  $G$  also acts on  $H^1(\Gamma, \mathcal{K}_{Z'}(V_n))$  and  $C_{U'}^1(V_n)$ . It is clear that the action of  $G$  commutes with the maps Frobenius action as well as the maps  $\iota'_{U',Z'}$ ,  $\gamma'_{U',Z'}$ ,  $P'_{U',Z'}$ ,  $T'_{U',Z'}$  and therefore also the monodromy operator  $N$  on  $H_{\text{dR}}^1(U', Z', \mathcal{V}_n)$ .

**Proposition 7.3.** *There is an exact sequence*

$$0 \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \xrightarrow{\iota'_{U,Z}} H_{\text{dR}}^1(U, Z, \mathcal{V}_n) \xrightarrow{\gamma'_{U,Z}} C_U^1(V_n)^\Gamma \rightarrow 0.$$

*Proof.* We begin with the analogous exact sequence for  $\Gamma'$  and take  $G$ -invariance to obtain

$$0 \rightarrow H^1(\Gamma, \mathcal{K}_{Z'}(V_n))^G \xrightarrow{\iota'_{U',Z'}} H_{\text{dR}}^1(U', Z', \mathcal{V}_n)^G \xrightarrow{\gamma'_{U',Z'}} (C_{U'}^1(V_n)^\Gamma)^G \rightarrow H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))).$$

Since  $G$  is a finite group, and  $H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n)))$  is a vector space over a field of characteristic zero, and hence is divisible, it follows that

$$H^1(G, H^1(\Gamma, \mathcal{K}_Z(V_n))) = 0.$$

We remark that there  $\mathcal{K}_{Z'} = \mathcal{K}_Z$  and  $C_U^1(V_n) = C_{U'}^1(V_n)$ . It follows by definition that

$$H_{\text{dR}}^1(U', Z', \mathcal{V}_n)^G = H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$$

and it is immediate that

$$(C_{U'}^1(V_n)^\Gamma)^G = C_U^1(V_n)^{\Gamma'}.$$

By the inflation-restriction sequence for group hypercohomology we obtain an exact sequence

$$0 \rightarrow H^1(G, H^0(\Gamma', \mathcal{K}_{Z'})) \rightarrow H^1(\Gamma, \mathcal{K}_Z(V_n)) \rightarrow H^1(\Gamma', \mathcal{K}_Z'(V_n))^G \rightarrow H^2(G, H^0(\Gamma', \mathcal{K}_{Z'}(V_n))).$$

Again, we note that  $G$  is a finite group and  $H^0(\Gamma', \mathcal{K}_{Z'}(V_n))$  is a divisible group, and so it follows that

$$H^1(\Gamma, \mathcal{K}_Z(V_n)) \cong H^1(\Gamma', \mathcal{K}_Z'(V_n))^G,$$

and the result holds.  $\square$

**Proposition 7.4.** *For  $\Gamma$  not necessarily torsion-free,  $H_{\text{dR}}^1(U, Z, \mathcal{V}_n)$  admits a  $(\varphi, N)$ -module structure that is independent of the choice of  $\Gamma'$ , and that there is a slope decomposition*

$$H_{\text{dR}}^1(U, Z, \mathcal{V}_n) = H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_m \oplus H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_{m+1},$$

where

$$H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_m \cong H^1(\Gamma, \mathcal{K}_Z(V_n)),$$

and

$$H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_{m+1} \cong C_U^1(V_n)^\Gamma.$$

There is also a decomposition

$$H_{\text{dR}}^1(U, Z, \mathcal{V}_n) = F_{U,Z}^{m+1} \oplus H_{\text{dR}}^1(U, Z, \mathcal{V}_n)_m,$$

where  $F_{U,Z}^{m+1}$  is the space of  $\Gamma$ -invariant meromorphic differentials on  $\mathcal{H}_p$  with poles supported on the set  $\bigcup_{i=1}^r \Gamma \cdot z_i$  modulo exact differentials.

**7.5. Poincaré Duality.** As before, let  $D_1 = (\tilde{z}_1) + \cdots + (\tilde{z}_r)$  and  $D_2 = (\tilde{w}_1) + \cdots + (\tilde{w}_s)$  be two divisors on  $X_\Gamma$ . Let  $U_j$  denote the open subscheme  $X_\Gamma - |D_j|$  for  $j \in \{1, 2\}$  and

$$i_j : Z_j \longrightarrow X_\Gamma$$

denote the closed subscheme associated to the divisor  $D_j$  for  $j \in \{1, 2\}$ . Let  $j, l \in \{1, 2\}$  be two distinct elements. We remark that the complex  $\Omega_{(U_j, Z_l)}^\bullet(\mathcal{V}_n)$ , whose hypercohomology computes  $H_{\text{dR}}^1(U_j, Z_l, \mathcal{V}_n)$ , is easily seen to be quasi-isomorphic to its subcomplex

$$[\mathcal{O}_{U_j}(-D_l) \otimes \mathcal{V}_n \rightarrow \Omega_{U_j}^1 \otimes \mathcal{V}_n].$$

It follows that  $H_{\text{dR}}^1(U_j, Z_l, \mathcal{V}_n)$  is computed as the hypercohomology of the complex

$$[\mathcal{O}_{X_\Gamma}(-D_l) \otimes \mathcal{V}_n \rightarrow \Omega_{X_\Gamma}^1(\log(D_j)) \otimes \mathcal{V}_n].$$

Using this complex, we have the usual cup product in de Rham cohomology

$$\langle \cdot, \cdot \rangle_{D_1, D_2} : H_{\text{dR}}^1(U_j, Z_l, \mathcal{V}_n) \times H_{\text{dR}}^1(U_l, Z_j, \mathcal{V}_n) \xrightarrow{\cup} H_{\text{dR}}^2(X_\Gamma) \xrightarrow{\text{tr}} K.$$

what is the field of definition to use here?.

**Proposition 7.5.** *The pairing  $\langle \cdot, \cdot \rangle_{D_1, D_2}$  is non-degenerate. Furthermore,  $F_{U_j, Z_l}^{m+1}$  and  $F_{U_l, Z_j}^{m+1}$  are orthogonal complements.*

*Proof.* The key input is Serre duality which implies that the cup products

$$H^1(X_\Gamma, \mathcal{O}_{X_\Gamma}(-D_j) \otimes \mathcal{V}_n) \times H^0(X_\Gamma, \Omega_{X_\Gamma}^1(\log(D_j)) \otimes \mathcal{V}_n) \xrightarrow{\cup} H^1(X_\Gamma, \Omega_{X_\Gamma}^1) \xrightarrow{\text{tr}} K$$

are non-degenerate for each  $j \in \{1, 2\}$ . The proof follows *mutatis mutandis* the proof of Poincaré duality. See [?]  $\square$

8. NEKOVÁŘ'S  $p$ -ADIC HEIGHTS

**8.1. Local  $p$ -adic heights of biextensions.** In this short section we recall the construction of Nekovář's local  $p$ -adic height pairing in the semistable case [Nek93].

Consider a biextension diagram of semistable  $p$ -adic representations of  $\text{Gal}(\bar{K}/K)$  for  $K/\mathbb{Q}_p$  finite extension

$$(8.1) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E_2 & \longrightarrow & V \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Q}_p(1) & \longrightarrow & E & \longrightarrow & E_1 \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \mathbb{Q}_p & = & \mathbb{Q}_p \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array} .$$

We assume that the following assumption holds for  $V$  and its twisted dual

$$(8.2) \quad D_{\text{cris}}(V)^{\varphi=1} = D_{\text{cris}}(V^*(1))^{\varphi=1} = 0.$$

The top short exact sequence induces a diagram (Sec. 3.5 [Nek93]) with exact rows and columns

$$(8.3) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ & & D_{\text{st}}(E_2)^{\varphi=1} & \xrightarrow{\sim} & D_{\text{st}}(V)^{\varphi=1} & & \\ & & \downarrow (0, N, -i) & & \downarrow (N, -i) & & \\ 0 & \longrightarrow & \mathbb{Q}_p \oplus K & \longrightarrow & \mathbb{Q}_p \oplus D_{\text{st}}(V(-1))^{\varphi=1} \oplus DR(E_2)/\text{Fil}^0 & \longrightarrow & D_{\text{st}}(V(-1))^{\varphi=1} \oplus DR(V)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H_{\text{st}}^1(K, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(K, E_2) & \longrightarrow & H_{\text{st}}^1(K, V) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

where we have denoted by  $i$  the inclusion map  $D_{\text{st}}(V) \hookrightarrow DR(V)$  and the same for  $E_2$ . Recall that this map depends on the choice of a branch of the  $p$ -adic logarithm.

Under the assumption (8.2), the embeddings of the  $DR$  components give isomorphisms

$$(8.4) \quad DR(E_2)/\text{Fil}^0(E_2) \cong \text{coker}(D_{\text{st}}(E_2)^{\varphi=1} \rightarrow D_{\text{st}}(V(-1))^{\varphi=1} \oplus DR(E_2)/\text{Fil}^0(E_2))$$

$$(8.5) \quad DR(V)/\text{Fil}^0(V) \cong \text{coker}(D_{\text{st}}(V)^{\varphi=1} \rightarrow D_{\text{st}}(V(-1))^{\varphi=1} \oplus DR(V)/\text{Fil}^0(V)).$$

This implies that from the exact diagram (8.3), it can be extracted the following commutative diagram with exact rows

$$(8.6) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Q}_p \oplus K & \longrightarrow & \mathbb{Q}_p \oplus DR(E_2)/\text{Fil}^0 & \longrightarrow & DR(V)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{st}}^1(K, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(K, E_2) & \longrightarrow & H_{\text{st}}^1(K, V) \longrightarrow 0 \end{array}$$

where the vertical isomorphisms are often referred to as *Bloch-Kato exponential maps*. This isomorphisms are explicitly given by Lemma 2.1 and Lemma 2.2. It is important to recall that the  $\mathbb{Q}_p$  summands are mapped isomorphically under the first horizontal map leaving with a commutative diagram of the form

$$(8.7) \quad \begin{array}{ccccccc} 0 & \longrightarrow & K & \longrightarrow & DR(E_2)/\text{Fil}^0 & \longrightarrow & DR(V)/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{f}}^1(K, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{f}}^1(K, E_2) & \longrightarrow & H_{\text{f}}^1(K, V) \longrightarrow 0. \end{array}$$

The definition of the *local  $p$ -adic height* will depend on the choice of the following two objects:

- a continuous homomorphism  $l : K^\times \rightarrow \mathbb{Q}_p$  admitting the following factorization

$$(8.8) \quad \begin{array}{ccc} \mathcal{O}_K^\times & \xrightarrow{l} & \mathbb{Q}_p \\ & \searrow \log \quad \nearrow \text{tr} & \\ & K & \end{array}$$

- a choice of a subspace  $W \subset DR(V)$  complementary to the filtered piece  $\text{Fil}^0$ , i.e. a decomposition  $DR(V) = W \oplus \text{Fil}^0(V)$ .

In particular, the choice of the subspace  $W$  induces a splitting of the top exact sequence as follows

$$(8.9) \quad w' : DR(V)/\text{Fil}^0(V) \xrightarrow{W} DR(V) \xrightarrow{\alpha} DR(E_2) \rightarrow DR(E_2)/\text{Fil}^0(E_2)$$

where the leftmost map is induced by the decomposition  $DR(V) = W \oplus \text{Fil}^0$  and  $\alpha$  denotes the unique  $(\varphi, N)$ -equivariant splitting

$$(8.10) \quad \begin{array}{ccccccc} 0 & \longrightarrow & K & \longrightarrow & DR(E_2)/\text{Fil}^0 & \xrightarrow{w'} & DR(V)/\text{Fil}^0 \longrightarrow 0. \end{array}$$

The splitting of the exact sequence induces a retraction in the bottom exact sequence

$$(8.11) \quad w : H_{\text{st}}^1(K, E_2) \rightarrow H_{\text{st}}^1(K, \mathbb{Q}_p(1)).$$

Recall that the *Kummer map* determines a canonical isomorphism

$$(8.12) \quad H_{\text{st}}^1(K, \mathbb{Q}_p) = K^\times \hat{\otimes} \mathbb{Q}_p.$$

Under this identification, the inverse of the left most vertical map of the diagram (8.10) is given by

$$(8.13) \quad \begin{array}{c} H_{\text{st}}^1(K, \mathbb{Q}_p(1)) \rightarrow \mathbb{Q}_p \oplus K \\ x \mapsto (v(x), \log(x)) \end{array}$$

for the given fixed choice of the branch of the  $p$ -adic logarithm. We are now ready to give the definition of the  $p$ -adic height.

**Definition 8.1.** The local  $p$ -adic height of the biextension  $[E] \in H_{\text{st}}^1(K, E_2)$  as in (8.1) is defined to be

$$(8.14) \quad h_{[E]} = -l(w([E])).$$

In the next section we will specialize this to the case of biextensions arising from Heegner cycles and we will provide an explicit formula for the local  $p$ -adic height.

**8.2. Heegner biextension.** We now want to study the extensions of filtered  $(\varphi, N)$ -modules obtained from the extensions of  $p$ -adic Galois representations considered in Section 4.

Let  $K$  be a number field unramified at  $p$ . Let  $D$  be a reduced effective divisor on  $X_M$  defined over  $K$  such that all closed points reduce to distinct points on  $X$  under the map  $q : X_M \rightarrow X$ , we have an extension

$$(8.15) \quad e_{\text{ét}} : 0 \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow H_{\text{ét}}^1((X_M - |D|) \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x(m) \rightarrow 0$$

applying the functor  $D_{\text{st}}$ , thanks to Theorem 5.3, we get an extension in  $\text{MF}_{K_p}(\varphi, N)$

$$(8.16) \quad e_{\text{dR}} : 0 \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \rightarrow H_{\text{dR}}^1(U, \mathcal{V}_n)[-m-1] \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m] \rightarrow 0.$$

Recall from (2.8), the relation between the effect of the Tate twist in the two categories. By Theorem 2.1, the equivalence of categories between semistable  $p$ -adic representations and admissible filtered  $(\varphi, N)$ -modules given by Fontaine's functor  $D_{\text{st}}$  is compatible with exact sequences, tensor product and duality. See in particular the discussion in Subsection 5.2 for the compatibility between the different pairings. Recall that from Subsection 4.4, the dual exact sequence  $e_{\text{ét}}^\vee$  is given by

$$(8.17) \quad e_{\text{ét}}^\vee : 0 \rightarrow \bigoplus_{x \in D} (\mathbb{L}_n)_x(m) \rightarrow H_{\text{ét}, |D|}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m))^G \rightarrow 0$$

while the corresponding dual sequence  $e_{\text{dR}}^\vee$  is given by

$$(8.18) \quad e_{\text{dR}}^\vee : 0 \rightarrow \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m] \rightarrow H_{\text{dR}}^1(X_\Gamma, D_\Gamma, \mathcal{V}_n)[-m] \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m] \rightarrow 0.$$

Considering two different reduced disjoint divisors  $D_1, D_2$  satisfying previous assumptions we can consider the biextension diagram constructed in Subsection 4.4 in the category  $\text{MF}_{K_p}(\varphi, N)$

$$(8.19) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m-1] & \longrightarrow & H_{\text{dR}}^1(U_{1,\Gamma}, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(U_{1,\Gamma}, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m] & \xlongequal{\quad} & \bigoplus_{z \in D_\Gamma} (\mathcal{V}_n)_z[-m] \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}.$$

We now want to consider only simple extensions of  $p$ -adic Galois representations by  $\mathbb{Q}_p$  and  $\mathbb{Q}_p(1)$  and the corresponding simple extensions of filtered  $(\varphi, N)$ -modules. In Section 6 we have recalled the construction of special Heegner classes  $\text{CH}^m(\mathcal{A}_P^m, \epsilon_n)$  attached to a Heegner point  $P$ . Under the  $p$ -adic Abel-Jacobi map, these classes can be used to produce extension classes

$$(8.20) \quad cl_0 : \text{CM}^{m+1}(\mathcal{M}_n) \rightarrow \text{Ext}_{\text{Rep}_{\text{st}}(G_K)}^1(\mathbb{Q}_p, H_{\text{ét}}^1(X_M \otimes \overline{\mathbb{Q}}, \mathbb{L}_n(m+1))^G)$$

Applying the functor  $D_{\text{st}, \mathbb{Q}_{p^2}}$  this produces a simple extension class in

$$(8.21) \quad \text{Ext}_{\text{MF}_{\mathbb{Q}_{p^2}}^{\text{ad}}(\varphi, N)}^1(\mathbb{Q}_{p^2}, H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]).$$

In [IS03], the authors explicitly characterized these latter extension classes arising from Heegner cycles.

Let  $P$  be a Heegner point on  $X$  defined over  $K$  quadratic imaginary number field of discriminant  $D_K$ . Let  $\Psi : K \rightarrow B$  be an optimal embedding corresponding to  $P$  as in Lemma 5.2. Let  $z_P \in \text{CH}^m(\mathcal{A}_P^m, \epsilon_n)$  be the Heegner class defined in Definition 6.2. Let  $P_\Psi(X, Y) \in \mathcal{P}_2$  the homogeneous polynomial in two variable defined by

$$(8.22) \quad P_\Psi(X, Y) = P_{\Psi(\sqrt{D_K})}(X, Y) = \text{tr} \left( \Psi_{\sqrt{D_K}} \begin{pmatrix} X \\ Y \end{pmatrix} \begin{pmatrix} X & -Y \end{pmatrix} \right)$$

as in the isomorphism described in Lemma 5.1



**Lemma 8.1.** *The composition of the  $p$ -adic Abel-Jacobi map*

$$(8.23) \quad cl : CH^m(\mathcal{A}_p^m, \epsilon_n) \rightarrow \mathbb{L}_{n,p}(m)$$

*with the canonical isomorphisms of  $(\varphi, N)$ -modules*

$$(8.24) \quad D_{st, \mathbb{Q}_{p^2}}(\mathbb{L}_n(m)_{\tilde{p}}) = (\mathcal{V}_n)_{\Psi}[-m]$$

*sends  $z_p^{(n)}$  to the element  $(-, P_{\Psi}^m) \in V_n = \mathcal{P}_n^{\vee}$ .*

*Proof.* See [IS03, Lemma 8.4] □

**8.3.  $p$ -adic heights as Poincaré pairing on de Rham cohomology.** In this section we will study how the  $p$ -adic pairing described in Subsection 8.1 can be explicitly computed in terms of the Poincaré pairing of special classes in de Rham cohomology.

We start by fixing  $D_1, D_2 \hookrightarrow X_{\Gamma}$  two disjoint effective reduced divisors on  $X_{\Gamma}$ . Assume that every closed point of  $D_1, D_2$  is defined over  $\mathbb{Q}_{p^2}$ . A simple extension class of  $H_{\text{dR}}^1(X_{\Gamma}, \mathcal{V}_n)[-m-1]$  by  $\mathbb{Q}_{p^2}$  can be constructed as the pullback along a map of filtered  $\varphi$ -modules  $cl_i : \mathbb{Q}_{p^2} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[-m]$  of the Gysin exact sequence

$$(8.25) \quad \begin{array}{ccccccc} 0 & \longrightarrow & H_{\text{dR}}^1(X_{\Gamma}, \mathcal{V}_n) & \longrightarrow & E_i & \longrightarrow & \mathbb{Q}_{p^2} \longrightarrow 0 \\ & & \parallel & & \downarrow \iota_{E_i} & & \downarrow cl_i \\ 0 & \longrightarrow & H_{\text{dR}}^1(X_{\Gamma}, \mathcal{V}_n) & \xrightarrow{\iota} & H_{\text{dR}}^1(X_{\Gamma} - D_i, \mathcal{V}_n)[-m-1] & \xrightarrow{\text{Res}} & \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[1] \longrightarrow 0. \end{array}$$

The notation  $cl$  is reminiscent of the fact that we will later construct such extensions from the Heegner classes previously described.

**Lemma 8.2.** *The exact sequence of filtered  $(\varphi, N)$ -modules*

$$(8.26) \quad 0 \longrightarrow H_{\text{dR}}^1(X_{\Gamma}, \mathcal{V}_n)[-m-1] \xrightarrow{\iota} H_{\text{dR}}^1(X_{\Gamma} - D_i, \mathcal{V}_n)[-m-1] \xrightarrow{\text{Res}} \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[-m] \longrightarrow 0.$$

*admits a unique Frobenius-monodromy equivariant splitting*

$$(8.27) \quad s_i : \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[1] \rightarrow H_{\text{dR}}^1(X_{\Gamma} - D_i, \mathcal{V}_n)$$

*where  $s_i(Q) = \omega_Q \in H_{\text{dR}}^1(X_{\Gamma} - D_i, \mathcal{V}_n)$  is the unique element satisfying*

- (i)  $\omega_Q \in H_{\text{dR}}^1(X_{\Gamma} - D_i, \mathcal{V}_n)_{m+1}$
- (ii)  $\text{Res}(\omega_Q) = Q$ ,
- (iii)  $N(\omega_Q) = 0$ .

*Proof.* First of all observe that  $\bigoplus_{z \in D_i} (\mathcal{V}_n)_z[1]$  is isotrivial of slope  $m+1$  since Frobenius acts by  $\det^{\otimes m}$  on  $V_n\{m\}$  and has trivial monodromy. In particular, Frobenius acts on every element  $Q$  as multiplication by  $p^{m+1}$ . See Subsection 5.1. [Cite the part on the decomposition of open cohomology](#)

in terms of group cohomology and harmonic cocycles. We have a commutative diagram of the form

$$(8.28) \quad \begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{I} & C_{\text{har}}(V_n)^\Gamma \longrightarrow 0 \\ & & \parallel & \swarrow P & \downarrow \iota & \searrow \tilde{\iota} & \\ 0 & \longrightarrow & H^1(\Gamma, V_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma - D, \mathcal{V}_n) & \xrightarrow{I_U} & C_U(V_n)^\Gamma \longrightarrow 0 \\ & & & \swarrow P_U & \downarrow \text{Res} & & \downarrow \\ & & & & \bigoplus_{z \in D} (\mathcal{V}_n)_z[1] & \longrightarrow & \bigoplus_{z \in D} \text{Ind}^\Gamma(V_n) \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & H^1(\Gamma, C_{\text{har}}(V_n)) \end{array}$$

change notation according to your notation in the section. From the diagram we then see that the intersection of the trivial monodromy  $N = 0$  and the slope  $m + 1$  cuts out a non trivial subspace. Cite this result from you section or add small argument. Suppose  $\omega, \omega'$  are two elements satisfying the previous properties then their difference lie in

$$\omega - \omega' \in \iota(H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1}) \cap \ker N = \{0\}$$

concluding the unicity by cite part on interaction between trivial monodromy and slope m+1.  $\square$

**Lemma 8.3.** Under the Bloch-Kato exponential map (2.10) we have an isomorphism

$$(8.29) \quad \text{Ext}_{\text{MF}_{\mathbb{Q}_{p^2}}^{\text{ad}}(\varphi, N)}^1(\mathbb{Q}_{p^2}, H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]) \cong (\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n))^\vee \cong M_k(X_\Gamma, \mathbb{Q}_{p^2})^\vee$$

and the extension class  $[E_i]$  of (8.25) is identified with the element  $(-, \eta_i) \in \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)^\vee$  where  $\eta_i \in H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$  is the uique element mod  $\text{Fil}^{m+1}$  satisfying

$$(8.30) \quad \iota(\eta_i) \equiv s_i(\text{cl}_i(1)) \mod \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

where  $s_i(\text{cl}_i(1)) \in H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$  is the element defined by the unique Frobenius-monodromy equivariant splitting of Lemma 8.2.

*Proof.* First of all, by the Bloch-Kato exponential the LHS of (8.29) is identified with  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)/\text{Fil}^{m+1}$  and by mention properties of Poincare duality and filtrations it is isomorphic to  $\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)^\vee$ . Furthermore we have the identification between filtered piece and modular forms thanks to Cite property of filtrations/modular forms contained in [IS03, Proposition 6.1].

Consider the pullback diagram (8.25). The construction in Lemma 8.2 induces a Frobenius-monodromy splitting of the top row of the diagram

$$(8.31) \quad \begin{aligned} s : \mathbb{Q}_{p^2} &\rightarrow E_i \\ 1 &\mapsto s_i(\text{cl}_i(1)). \end{aligned}$$

From the explicit construction of the Bloch-Kato discussed after Lemma 2.1, we have that  $\iota$  induces an isomorphism

$$(8.32) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)/\text{Fil}^{m+1} \cong E_i/\text{Fil}^{m+1} E_i$$

and  $\eta_i$  is defined to be the corresponding element in  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$  such that

$$(8.33) \quad \iota(\eta_i) \equiv s_i(\text{cl}_i(1)) \mod \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

concluding the description of the extension class  $[E_i]$  under the isomorphism (8.29).  $\square$

Similarly, we can describe the extension class of the middle column of (8.19). Fix two classes  $\text{cl}_i : \mathbb{Q}_{p^2} \rightarrow \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[-m]$ , let  $Q_1, Q_2$  the respective images of the element 1. Consider the dual shifted extension of  $E_2$  that can be obtained as the push-out along the cl map

(8.34)

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \bigoplus_{z \in D_2} (\mathcal{V}_n)_z & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \parallel \\
 0 & \longrightarrow & \mathbb{Q}_{p^2}[-1] & \longrightarrow & E_2^\vee[-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0
 \end{array}$$

In the previous top exact sequence we have used the fact that [add reference to cup/product duality between open/relative](#). Construct the biextension obtained from the two classes  $\text{cl}_i$  pulling back and pushing out the diagram (8.19)

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \mathbb{Q}_{p^2}[-1] & \longrightarrow & E_2^\vee[-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow \\
 (8.35) \quad 0 & \longrightarrow & \mathbb{Q}_{p^2}[-1] & \longrightarrow & E & \longrightarrow & E_1 \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & \mathbb{Q}_{p^2} & \xlongequal{\quad} & \mathbb{Q}_{p^2} \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

Lemma 2.2 explicitly identifies the extension class  $[E]$  as an element in

$$(8.36) \quad \text{Ext}_{\text{MF}_{\mathbb{Q}_{p^2}}^{\text{ad}}(\varphi, N)}^1(\mathbb{Q}_{p^2}, E_2^\vee[-1]) \cong \mathbb{Q}_{p^2} \oplus E_2^\vee[-1]/\text{Fil}^0(E_2^\vee[-1])$$

where  $E_2[-1]$  satisfies all of the assumptions of Lemma 2.2. Observe that under the Poincaré dual [cite relevant subsection for pairing in de rham cohomology](#) we can identify the group of extensions with

$$(8.37) \quad E_2^\vee[-1]/\text{Fil}^0(E_2^\vee[-1]) \cong (\text{Fil}^0 E_2)^\vee.$$

The extension class corresponding to  $E$  in (8.35) can be determined as follows.

**Lemma 8.4.** *Let  $\iota_{E_2}$  be the injection of  $E_2$  in  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  (8.25) and  $\iota$  be the injection of  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  in  $H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ . The element  $[E]$  in the extension group of  $E_2^\vee[-1]$  by  $\mathbb{Q}_{p^2}$  is given by*

$$(8.38) \quad (N_E, (\eta_E, \iota_{E_2}(-))) \in \mathbb{Q}_{p^2} \oplus (\text{Fil}^0 E_2)^\vee$$

where  $N_E$  is a certain element in  $\mathbb{Q}_{p^2}$ , and  $\eta_E \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  satisfies the following properties

- (i)  $\iota(\eta_E) \equiv \omega_{Q_1} \pmod{\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)}$  for  $\omega_{Q_1} \in H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$ ,
- (ii)  $\text{Res}(\omega_{Q_1}) = Q_1$ ,
- (iii)  $\varphi(\omega_{Q_1}) = p^{m+1} \omega_{Q_1}$ . [We need to discuss this condition](#)

The cohomology class  $\omega_{Q_1}$  is uniquely characterized by the property (ii), (iii).

*Proof.* This time we need to apply Lemma 2.2, the general version of Lemma 2.1. First of all, we consider the pull-back of the last two columns of the biextension diagram (8.19) along the map

$\text{cl}_i : \mathbb{Q}_{p^2} \hookrightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[-m]$  obtaining the following diagram.

$$(8.39) \quad \begin{array}{ccccc} & & 0 & & 0 \\ & & \downarrow & & \downarrow \\ & & 0 & & 0 \\ & & \downarrow & & \downarrow \\ H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \\ & \searrow & \downarrow & \searrow & \downarrow \\ & & H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \\ & & \downarrow & & \downarrow \\ H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)[-m-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)[-m-1] \\ & \searrow & \downarrow & \searrow & \downarrow \\ & & E' & & E_1 \\ & & \downarrow & & \downarrow \\ \oplus_{z \in D_1} (\mathcal{V}_n)_z[-m] & \xrightarrow{\quad} & \oplus_{z \in D_1} (\mathcal{V}_n)_z[-m] & \xrightarrow{\quad} & \oplus_{z \in D_1} (\mathcal{V}_n)_z[-m] \\ & \searrow & \downarrow & \searrow & \downarrow \\ & & \mathbb{Q}_{p^2} & & \mathbb{Q}_{p^2} \\ & & \downarrow & & \downarrow \\ & & 0 & & 0 \end{array}$$

As defined in Lemma 8.2, we have a Frobenius-monodromy equivariant splitting of the extension corresponding to  $E_1$  given by (8.27). We cannot hope to define such a splitting for the extension corresponding to  $E'$  since the  $(\varphi, N)$ -module  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$  contains a non-trivial submodule with trivial monodromy  $N = 0$ . In fact,  $H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)[-m-1]$  is obtained as an extension of  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]$  by  $\bigoplus_{z \in D_2} (\mathcal{V}_n)_z[-m-1]$ . As in Lemma 2.2 and in the discussion that follows, we construct a  $\varphi$ -equivariant splitting of the surjective map  $\text{Res}$

$$(8.40) \quad s : \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[1] \rightarrow H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$$

that we can describe as follows. Observe that  $\bigoplus_{z \in D_1} (\mathcal{V}_n)_z[1]$  is isotrivial of slope  $m+1$  since Frobenius acts by  $\det^{\otimes m}$  on  $\mathcal{V}_n\{m\}$  and trivial monodromy. See Subsection 5.1. For every  $Q \in \bigoplus_{z \in D_1} (\mathcal{V}_n)_z[1]$ , define  $s(Q) \in H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)$  to be the class  $\omega_Q$  satisfying

- $\text{Res}(\omega_Q) = Q$ ,
- $\varphi(\omega_Q) = p^{m+1}\omega_Q$ .

Such an element is unique because if  $\omega, \omega'$  are two elements satisfying such properties, then their difference lie in

$$\omega - \omega' \in \iota(H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_{m+1})^{\varphi=p^{m+1}}.$$

Observe that as  $\varphi$ -modules we have a decomposition

$$(8.41) \quad H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n) = H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \oplus \bigoplus_{z \in D_2} (\mathcal{V}_n)_z$$

and in particular

$$(8.42) \quad H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)_{m+1} = H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1}$$

concluding the unicity by then noting that the

$$(8.43) \quad (H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_{m+1})^{\varphi=p^{m+1}} = \{0\}.$$

By the commutativity of the diagram (8.39) and the compatibility of the Frobenius equivariant splitting, we have that the map

$$(8.44) \quad H_{\text{dR}}^1(X_\Gamma - D_1, D_2, \mathcal{V}_n)[-m-1] \rightarrow H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)[-m-1]$$

sends  $s(\text{cl}_1(1))$  to the cohomology class  $\omega$  satisfying

- (i)  $\omega \in H_{\text{dR}}^1(X_\Gamma - D_1, \mathcal{V}_n)_{m+1}$
- (ii)  $\text{Res}(\omega) = Q_1$ ,
- (iii)  $N(\omega) = 0$

that is the image of 1 under the Frobenius-monodromy equivariant map (8.31). Consider now the push-out of (8.39) along the map  $\text{cl}_2^\vee : \bigoplus_{z \in D_2} (\mathcal{V}_n)_z[-m-1] \rightarrow \mathbb{Q}_{p^2}[-1]$

(8.45)

$$\begin{array}{ccccccc}
 0 \rightarrow \bigoplus_{z \in D_2} (\mathcal{V}_n)_z[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] & \longrightarrow & 0 \\
 & \searrow & \downarrow & \searrow & \downarrow & \searrow & \\
 0 & \xrightarrow{\quad} & \mathbb{Q}_{p^2}[-1] & \xrightarrow{\quad} & E_2^\vee[-1] & \xrightarrow{\quad} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \rightarrow 0 \\
 & \downarrow & \downarrow & & \downarrow & & \downarrow \\
 0 \rightarrow \bigoplus_{z \in D_2} (\mathcal{V}_n)_z[-m-1] & \xrightarrow{\quad} & E' & \xrightarrow{\quad} & E_1 & \xrightarrow{\quad} & 0 \\
 & \searrow & \downarrow & \searrow & \downarrow & \searrow & \\
 0 & \xrightarrow{\quad} & \mathbb{Q}_{p^2}[-1] & \xrightarrow{\quad} & E & \xrightarrow{\quad} & E_1 \longrightarrow 0
 \end{array}$$

Observe that we have a Frobenius stable decomposition of  $E$  that is given by

$$(8.46) \quad E = \mathbb{Q}_{p^2}[-1] \oplus H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1] \oplus \mathbb{Q}_{p^2}$$

and the element  $s(\text{cl}_1(1))$  previously considered lands in the  $\mathbb{Q}_{p^2}$ -part. We now observe that under the inclusion

$$(8.47) \quad \iota : H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \rightarrow E'$$

we have an isomorphism of  $\mathbb{Q}_{p^2}$ -vector spaces given by

$$(8.48) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)/\text{Fil}^{m+1} \cong E'/\text{Fil}^0(E')$$

that means that there exists a class  $\eta_E \in H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)$  satisfying the following property

$$(8.49) \quad \iota(\eta_E) \equiv s(\text{cl}_1(1)) \pmod{\text{Fil}^0(E')}.$$

We can then define  $\tilde{\eta}_E$  to be the image of  $\eta_E$  under the surjection map

$$(8.50) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)/\text{Fil}^{m+1} \rightarrow E_2^\vee[-1]/\text{Fil}^0(E_2^\vee[-1]).$$

The RHS of the last equation is isomorphic to  $(\text{Fil}^{m+1}E_2)^\vee$  under Poincaré duality. Recall that  $E_2$  injects into

$$(8.51) \quad \iota_{E_2} : E_2 \hookrightarrow H_{\text{dR}}^1(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m-1]$$

as filtered  $(\varphi, N)$ -modules. Recall that we have a Poincaré pairing [cite relevant section where Poincare pairing is defined](#)

$$(8.52) \quad H_{\text{dR}}^1(X_\Gamma, D_{2,\Gamma}, \mathcal{V}_n)[-m-1] \otimes H_{\text{dR}}^1(X_\Gamma - D_{2,\Gamma}, \mathcal{V}_n)[-m] \rightarrow \mathbb{Q}_{p^2}$$

This implies that element  $\tilde{\eta}_E$  can be identified with the functional

$$(8.53) \quad \tilde{\eta}_E = (\tilde{\eta}_E, -) = (\eta_E, \iota_{E_2}(-)) \in (\text{Fil}^0 E_2)^\vee.$$

Under the Bloch-Kato exponential map we can then identify the class  $[E]$  with

$$(8.54) \quad [E] = (N(s(\text{cl}_1(1))), (\eta_E, \iota_{E_2}(-))) \in \mathbb{Q}_{p^2} \oplus (\text{Fil}^0 E_2)^\vee$$

concluding the proof.  $\square$

Recall from Proposition 7.1 that we have a decomposition of the de Rham cohomology by

$$(8.55) \quad H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) = \text{Fil}^{m+1} \oplus H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_m.$$

This will be our choice of decomposition of the de Rham realization needed to define the  $p$ -adic height pairing.

**Definition 8.2.** For every  $Q \in \bigoplus_{z \in D_i} (\mathcal{V}_n)_z$  define the Coleman-Gross differential associated to  $Q$  to be the unique class

$$(8.56) \quad \Psi_Q \in \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n)$$

such that  $\text{Res}(\Psi_Q) = Q$  and its image under the Frobenius equivariant splitting  $H_{\text{dR}}^1(X_\Gamma - D_i, \mathcal{V}_n) \rightarrow H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$  is contained in  $H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)_m$

**Theorem 8.1.** The local  $p$ -adic height pairing of the biextension  $[E]$  obtained from the class maps  $\text{cl}_i : \mathbb{Q}_{p^2} \rightarrow \bigoplus_{z \in D_i} (\mathcal{V}_n)_z[1]$  is given by

$$(8.57) \quad h_{[E]} = -\text{tr}(\eta_E, \Psi_{Q_2})$$

where  $\eta_E \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  is the class described in Lemma 8.4.

*Proof.* Recall from Subsection 8.1 that the local  $p$ -adic height of the class of a biextension  $[E] \in H_{\text{st}}^1(\mathbb{Q}_{p^2}, E_2^\vee[-1])$  is defined by

$$(8.58) \quad h_{[E]} = -l_v(w([E]))$$

where  $w$  is the splitting of the bottom exact sequence

$$(8.59) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Q}_{p^2} \oplus \mathbb{Q}_{p^2} & \longrightarrow & \mathbb{Q}_{p^2} \oplus DR(E_2^\vee[-1])/\text{Fil}^0 & \longrightarrow & DR(H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1])/\text{Fil}^0 \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & \swarrow w' & \downarrow \sim \\ 0 & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, \mathbb{Q}_p(1)) & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, E_2^\vee[-1]) & \longrightarrow & H_{\text{st}}^1(\mathbb{Q}_{p^2}, H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)[-m-1]) \longrightarrow 0. \\ & & \swarrow w & & & & \end{array}$$

The map  $w$  is induced by an explicit splitting of the upper exact sequence. Recall that  $\mathbb{Q}_{p^2}$  is mapped isomorphically to itself through the first horizontal map (8.7). Furthermore, we can identify under Poincaré duality the quotients in the upper exact sequence as

$$(8.60) \quad 0 \longrightarrow \mathbb{Q}_{p^2} \longrightarrow (\text{Fil}^0 E_2)^\vee \longrightarrow (\text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n))^\vee \longrightarrow 0$$

inducing a dual exact sequence contained in the usual Gysin extension

$$(8.61) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \text{Fil}^0 E_2 & \longrightarrow & \mathbb{Q}_{p^2} \longrightarrow 0 \\ & & \parallel & \swarrow w^{\vee} & \downarrow \iota_{E_2} & & \downarrow \text{cl}_2 \\ 0 & \longrightarrow & \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \xrightarrow{\iota} & \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \xrightarrow{\text{Res}} & \bigoplus_{z \in D_2} \text{Fil}^m(\mathcal{V}_n)_z \longrightarrow 0. \end{array}$$

By Lemma 8.4 we have that the class of  $[E]$  under the Bloch-Kato isomorphism is send to

$$(8.62) \quad (N_E, (\eta_E, \iota_{E_2}(-))) \in \mathbb{Q}_{p^2} \oplus \text{Fil}^0 E_2$$

with  $\eta_E \in H_{\text{dR}}^1(X_\Gamma, D_2, \mathcal{V}_n)$  satisfying prescribed properties of the Lemma. In Subsection 8.1 we have described a section of the top exact sequence that Nekovář constructs in [Nek93]. Explicitly this is given by

$$(8.63) \quad w^{\vee} : \text{Fil}^0 E_2 \longrightarrow E_2 \xrightarrow{\alpha} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \longrightarrow \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n)$$

where the first map is the canonical inclusion,  $\alpha$  is the unique  $(\varphi, N)$ -equivariant splitting of the canonical inclusion and the last morphism it the projection according to the decomposition (8.55). This map exists in more generality as the composition of the equivalent maps

$$(8.64) \quad \begin{array}{ccccccc} \text{Fil}^0 E_2 & \longrightarrow & E_2 & \xrightarrow{\alpha} & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) \\ \downarrow & & \downarrow & & \parallel & & \parallel \\ \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma - D_2, \mathcal{V}_n) & \longrightarrow & H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n) & \longrightarrow & \text{Fil}^{m+1} H_{\text{dR}}^1(X_\Gamma, \mathcal{V}_n). \end{array}$$

The Frobenius-monodromy equivariant splitting is given by Lemma 8.2. Let  $\omega_{Q_2} = s_2(Q_2)$  be the image of the element  $Q_2$  under the  $(\phi, N)$ -equivariant splitting. The splitting of the dual filtered exact sequence (8.61) can be then given as

$$(8.65) \quad \mathbb{Q}_{p^2} \rightarrow \mathrm{Fil}^0 E_2$$

$$(8.66) \quad 1 \mapsto \Psi_{Q_2} = \omega_{Q_2} - (\iota \circ w^{\vee})(\omega_{Q_2})$$

where  $\Psi_{Q_2}$  is then Coleman-Gross differential associated to  $Q_2$ . We can then see that the splitting of the exact sequence (8.59) is given by

$$(8.67) \quad \mathbb{Q}_{p^2} \oplus DR(E_2^{\vee}[-1])/\mathrm{Fil}^0 \rightarrow \mathbb{Q}_{p^2} \oplus \mathbb{Q}_{p^2} \\ (n, (\omega, \iota_{E_2}(-))) \mapsto (n, (\omega, \Psi_{Q_2})).$$

Under assumption (8.8)  $l = \mathrm{tr} \circ \log$  and the Kummer isomorphism (8.12), we conclude that the  $p$ -adic height is given by the projection of the second component of (8.67) composed with the trace map. This concludes the proof.  $\square$

**Corollary 8.1.** *The local  $p$ -adic height of the biextension  $[E]$  can be computed by*

$$(8.68) \quad h_{[E]} = -\mathrm{tr} \left( \int_{D_1} \Psi_{Q_2} Q_1 dz \right)$$

*Proof.*  $\square$

## 9. HIGHER $p$ -ADIC GREEN FUNCTIONS

### 9.1. Poincaré series.

### 9.2. An explicit formula.

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